

Cooperative Localization and Control for Heterogeneous Mobile Robots

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Abstract

- This thesis contributes to the UTOPIA project, aiming to develop cooperative localization and control strategies for heterogeneous mobile robots in challenging environments like agriculture.
- It involves designing a UAV and a UGV with unique sensors and maneuverability.
- The EKF was explored in different configurations to enhance each robot's localization precision.
- The EKF's performance was assessed in three cooperative missions, in simulations and real-world situations.

UTOPIA Project

- This international project develops an open framework to improve precision agriculture.
- It aims to create a cloud-based repository for related data, with an easy-to-use interface.
- The framework allows using robotic platforms like UAVs, USVs, and UGVs for stakeholder-specified missions, saving labor and time.

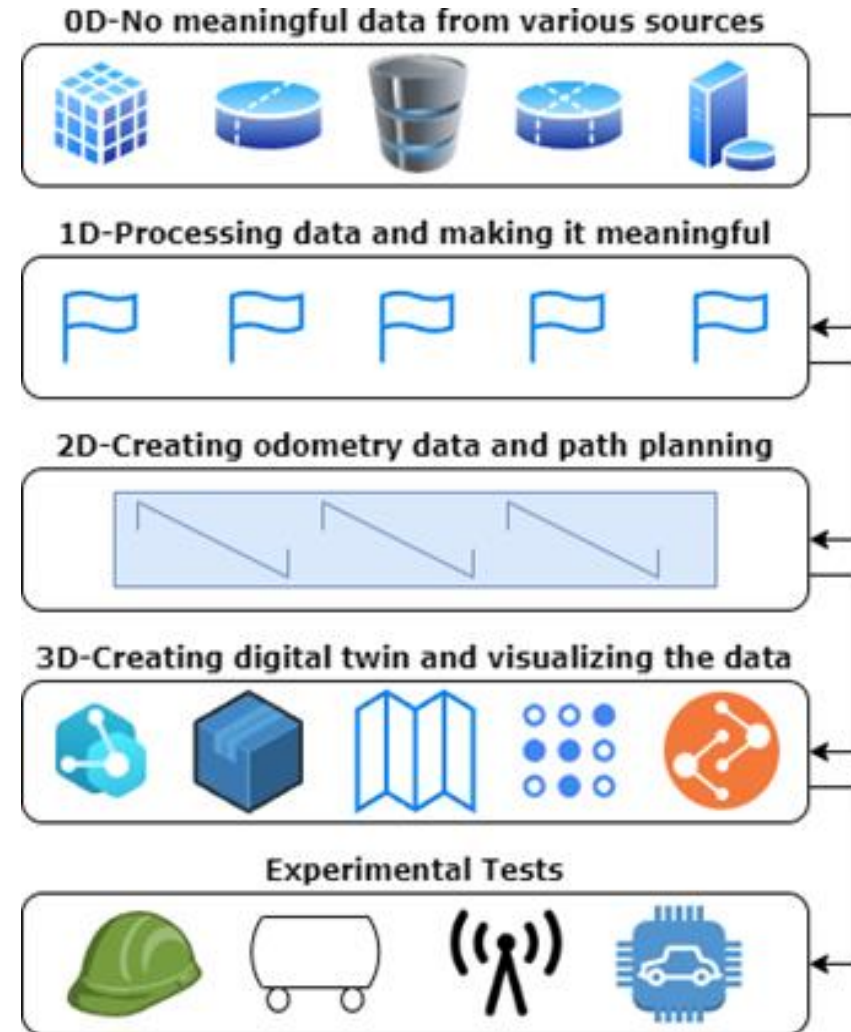


The figure above shows the field where UTOPIA's vineyard suitcase tests are conducted.

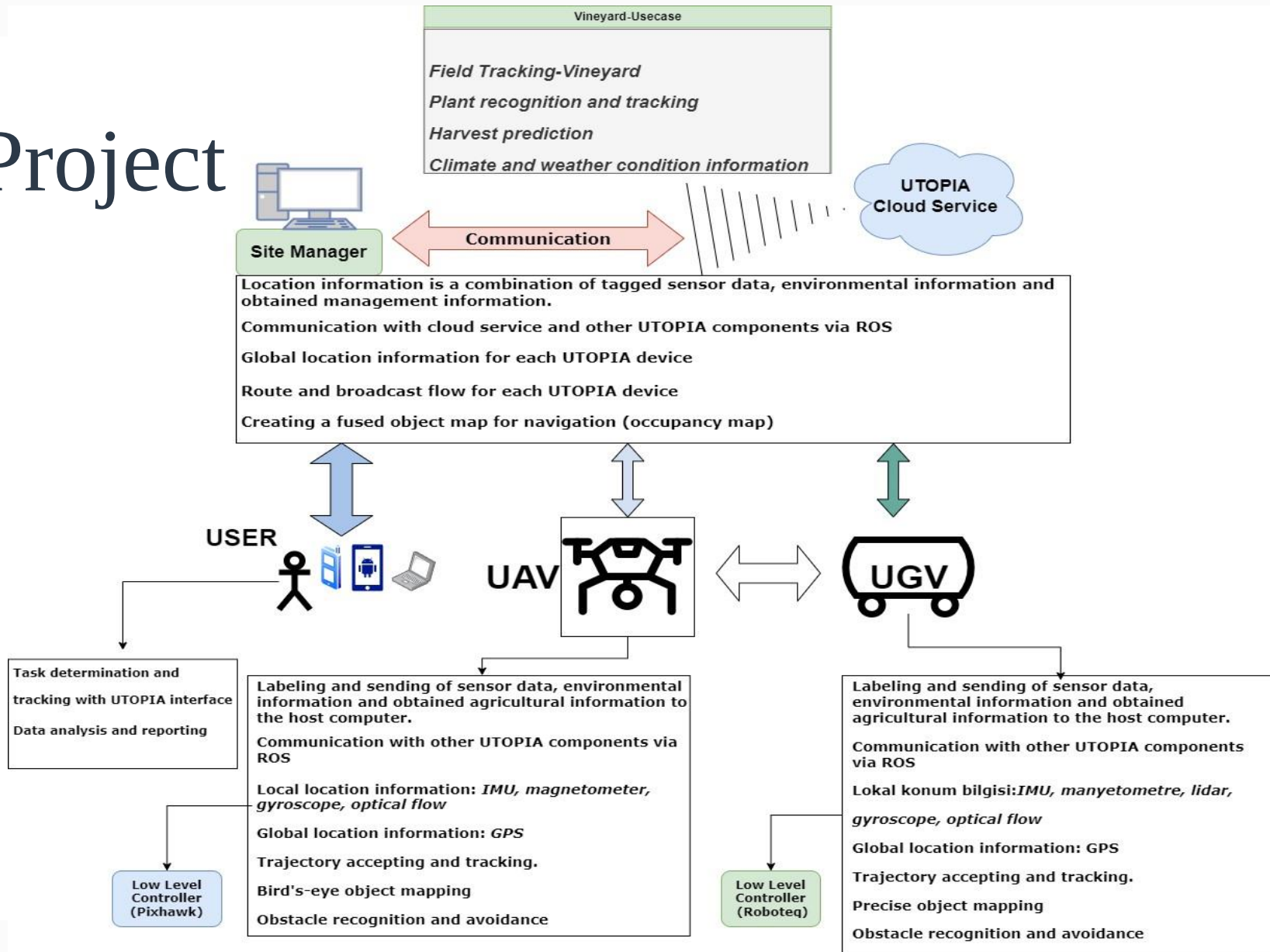


UTOPIA Project

- The nearby figure depicts UTOPIA devices' structure, demonstrating data's journey to influence real-world experiments.
- Initially, context-less data streams are collected. «
- This data is converted into meaningful information through processing, forming a 1D representation.
- Insights from this processed data generate 2D path planning and odometry data for spatial navigation and trajectory planning.
- Finally, these 2D representations evolve into a 3D digital twin, visualizing the operational environment.



UTOPIA Project



Motivation

- This thesis addresses the challenges in outdoor mobile robot applications in agriculture.
- It proposes the integration of diverse robots like UAVs and UGVs to enhance cooperative missions.
- The focus is on how these robots can collaboratively increase versatility and efficiency in cooperative missions with a robust localization and control.

Motivation

A brief comparison for UAV and UGV is given in the table.

Aspect	UAV	UGV
Advantages	Aerial perspective for reconnaissance and surveillance.	Provides a stable platform for carrying heavy payloads.
	Rapid deployment and mobility over obstacles and rough terrain.	Offers greater endurance and longer operating times compared to UAVs.
	Ability to cover large areas quickly.	Has a lower risk of collisions and damage in cluttered environments.
	No ground contact, which reduces wear and tear.	Can operate in GPS-denied or indoor environments.
Disadvantages	Limited payload capacity compared to UGVs.	Limited mobility in complex or vertical environments.
	Restricted flight time due to battery limitations.	Slower speed compared to UAVs, especially over long distances.
	Vulnerability to adverse weather conditions, such as wind and rain.	Susceptibility to terrain constraints and obstacles.
	Requirement for airspace regulation compliance.	Limited line-of-sight perception compared to UAVs.

Problem Statement and Objectives

- Integrating a quadcopter as UAV and a differential drive robot as UGV provides a potential solution for outdoor use.
- Quadcopters offer maneuverability and versatility for aerial tasks, but require robust control and precision localization.
- UGVs excel in ground tasks but need effective localization for accurate navigation.
- Both platforms need robust localization and control systems to optimize their benefits.

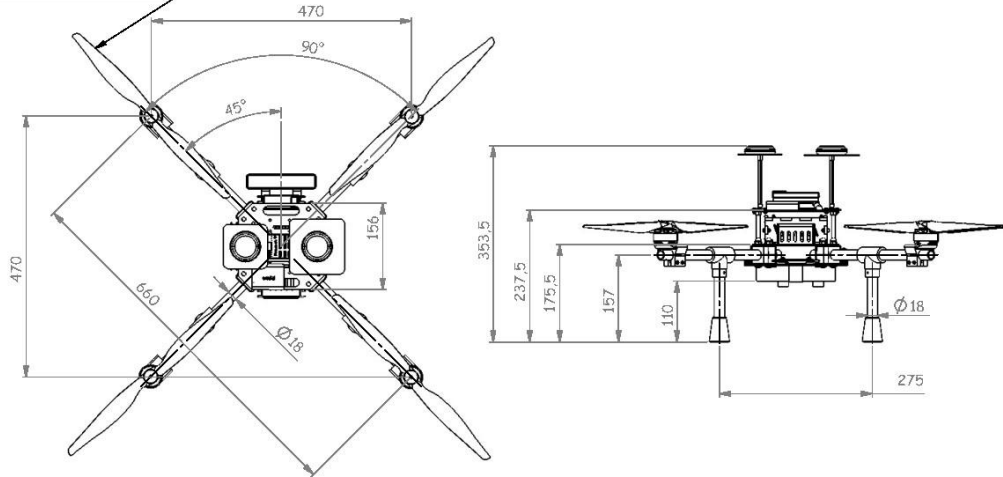
Literature Survey

- Research on cooperative localization and control for diverse mobile robots is infrequent, indicating scope for further study.
- Previous research largely targets multi-robot systems, not specifically diverse mobile robots, and mostly in indoor settings.
- This thesis explores both cooperative control and localization for diverse mobile robots, especially in demanding outdoor areas like agricultural fields.

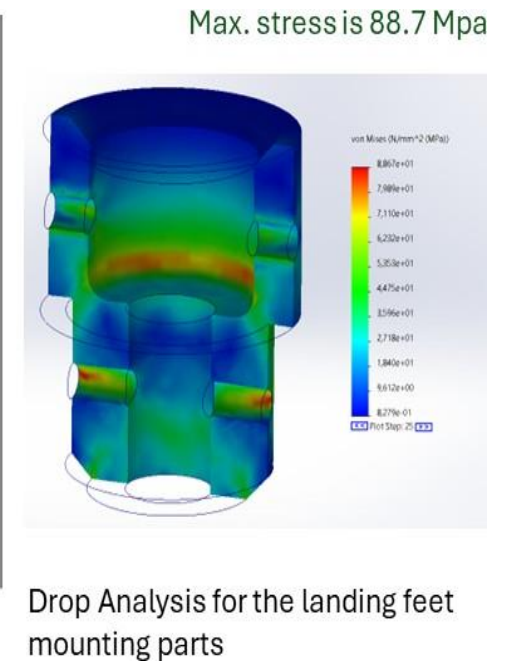
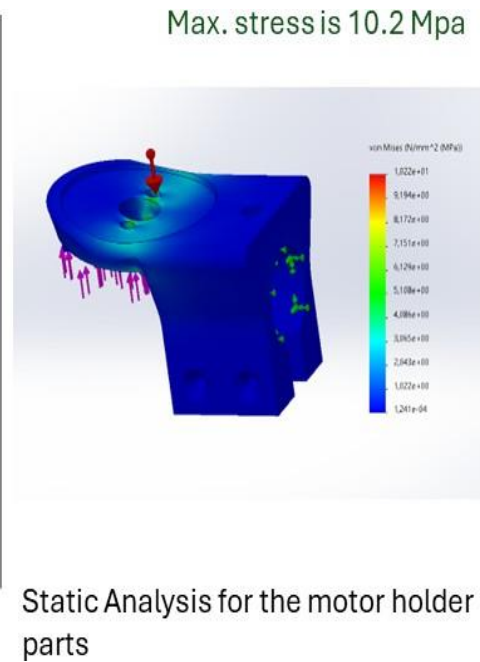
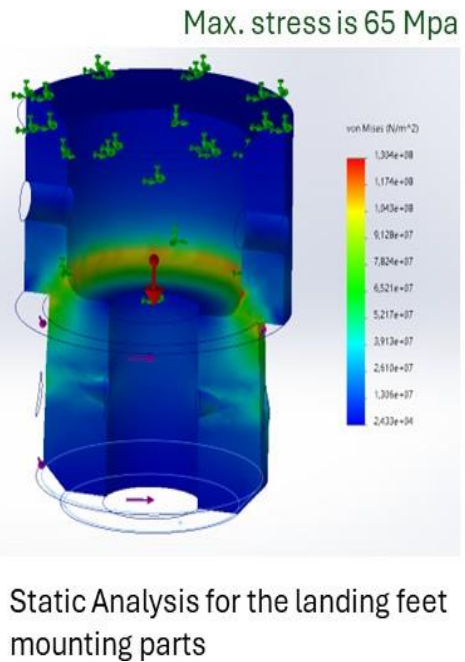
Hardware Design-The UAV

- The UAV used in this thesis is a carefully designed and built X-configured quadcopter.
- Its design was thoroughly verified through static and flow analysis to ensure structural integrity and top performance.

16x8" Carbonfiber Propellers

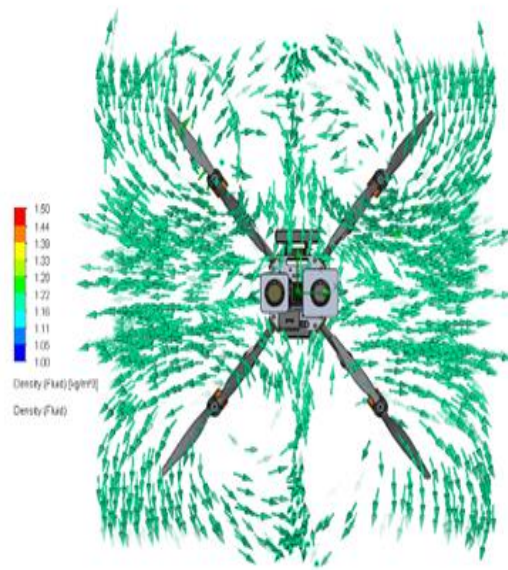


Hardware Design-The UAV

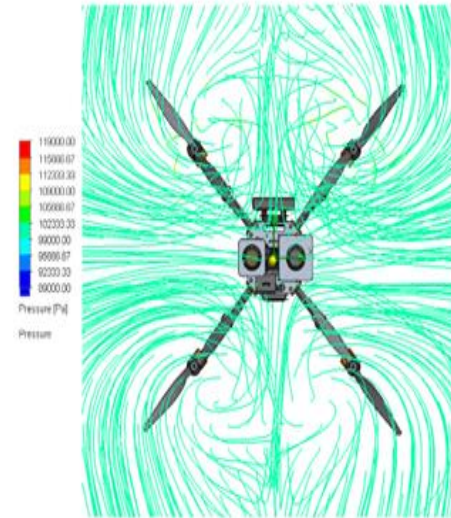


The UAV weighs 3.8 Kg
 Total thrust is $4 \times 1.8 = 7.2$ Kg
 The minimum required thrust is $3.8 \times 1.8 = 6.8$ Kg

Hardware Design-The UAV



Flow Density



Flow Pressure

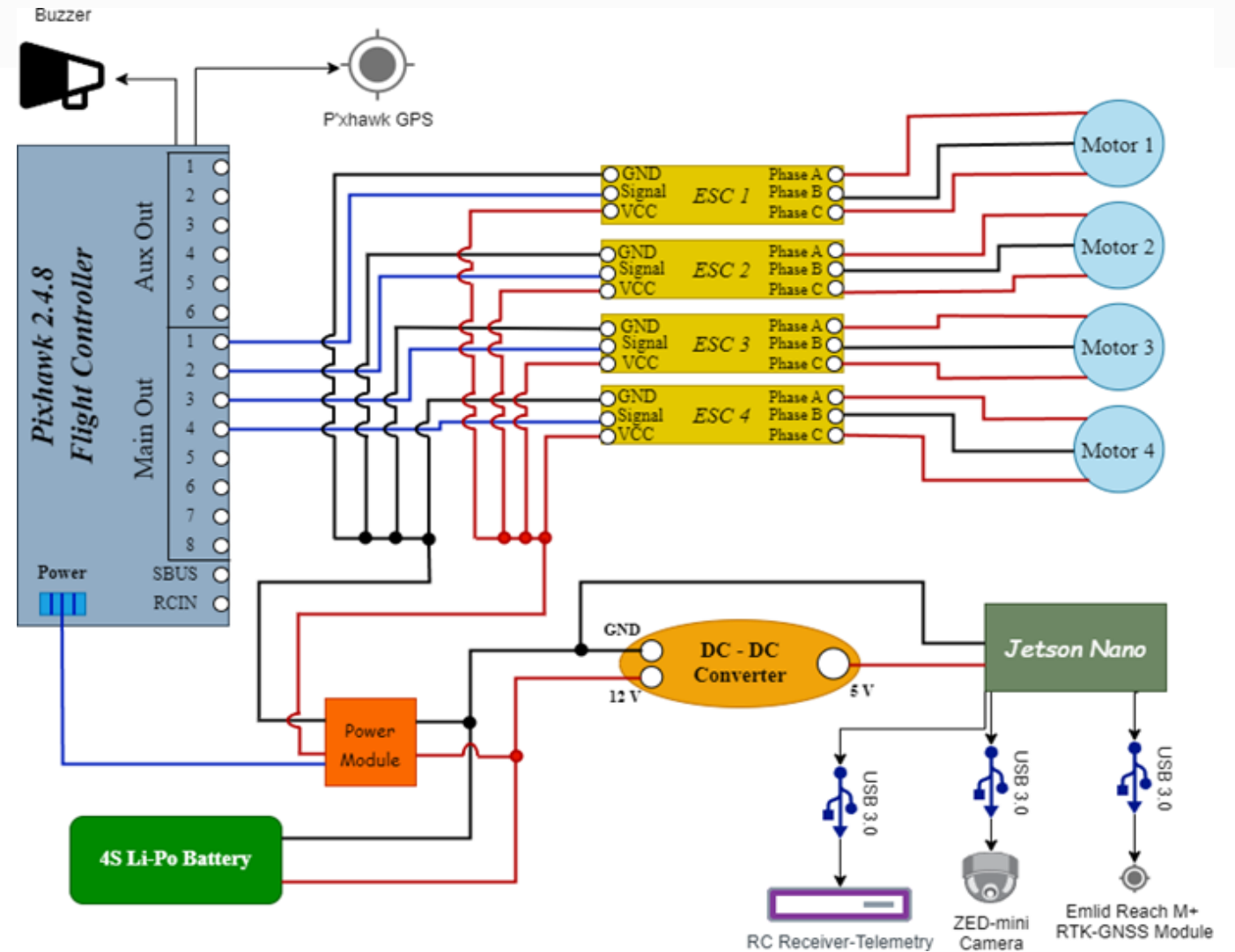
Total impact is 6.8 N



Drop Analysis for the landing feet mounting parts

Hardware Design- The UAV

No.	Component	Quantity
1	4S Li-Po Battery	1
2	Adjustable 5V DC-DC Converter	1
3	Buzzer	1
4	Brushless DC Motor (T-Motor U3)	4
5	Emlid Reach M+ RTK-GNSS Module	1
6	ESC 30A	4
7	Nvidia Jetson Nano Controller Board	1
8	Pixhawk 2.4.8 Flight Controller Board	1
9	Radio Controller Receiver and Transmitter	1
10	RadioLink Telemetry	1
11	ZED-mini Stereo Camera	1



Hardware Design-The UAV



ZEDmini Camera



Emlid Reach
M+ RTK GNSS
Modules
Base and Rover



RC Transmitter



Pixhawk 2.4.8 Flight Controller
and Telemetry & RC Receiver



DC-DC
Converter



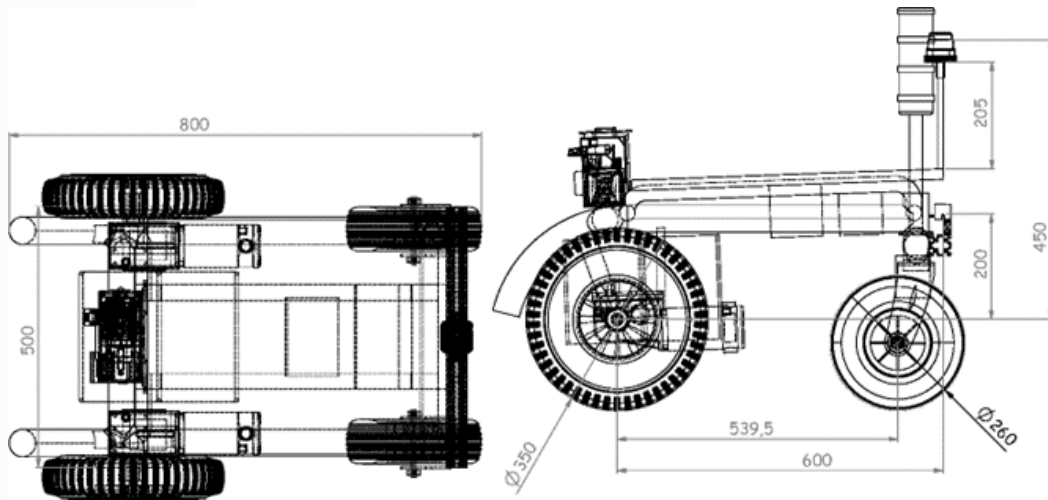
Jetson Nano Developer Board

Hardware Design-The UAV



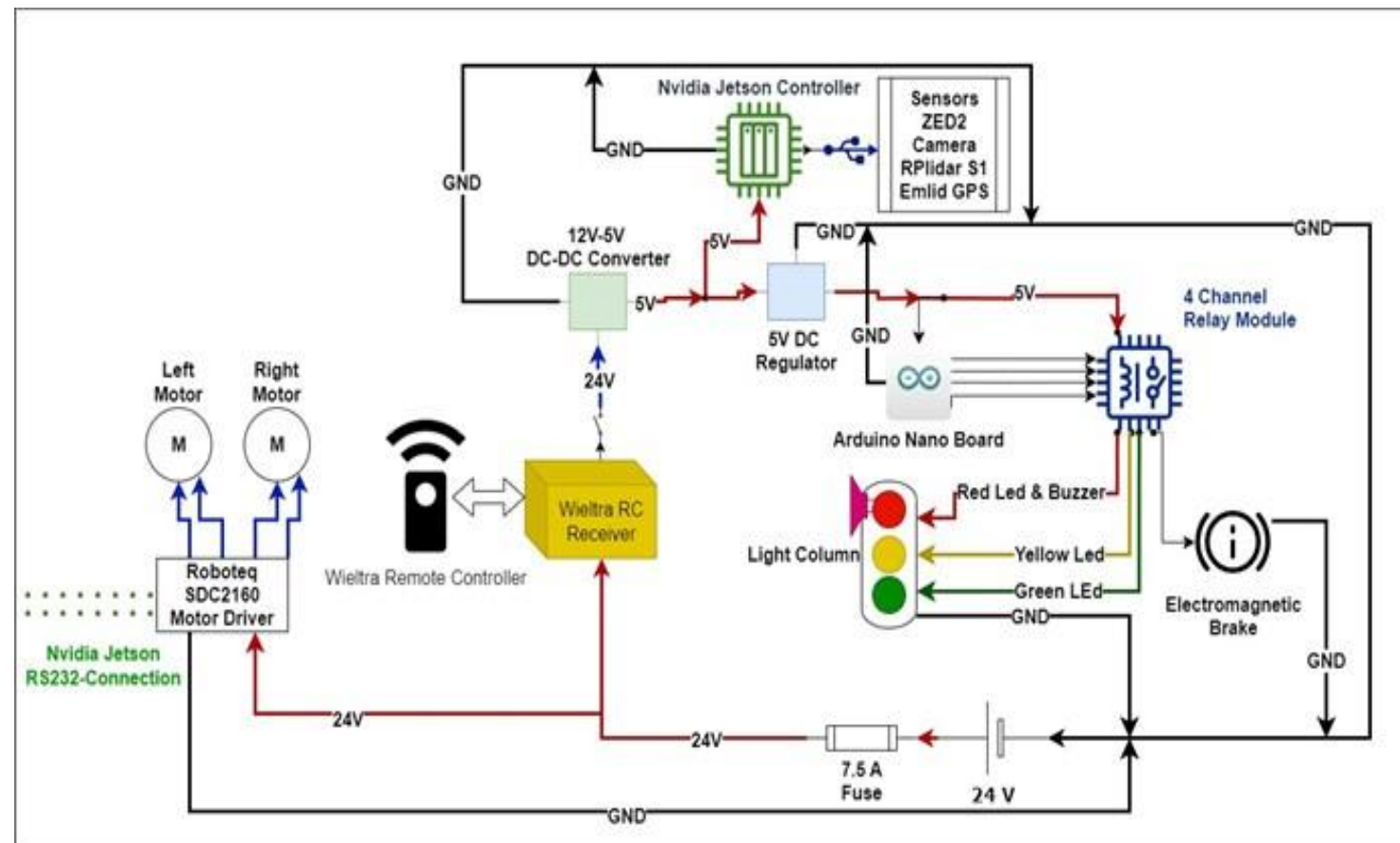
Hardware Design-The UGV

- The UGV used in this thesis is built from an electric wheelchair chassis.
- This two-wheeled differential drive vehicle provides a robust base for autonomous navigation tasks.



Hardware Design- The UGV

No.	Component	Quantity
1	12V 40Ah Battery	2
2	4-Channel Relay Module	1
3	Adjustable 5V DC-DC Converter	1
4	Arduino Nano Board	1
5	Brushed DC Motor & 32:1 Gearbox	2
6	Emlid Reach M+ RTK-GNSS Module	1
7	Light Column	1
8	Nvidia Jetson Nano Controller Board	1
9	Roboteq SDC2160 Motor Driver	4
10	RPLidar S1	1
11	Wieltra RC Transmitter & Receiver	1
12	ZED2 Stereo Camera	1

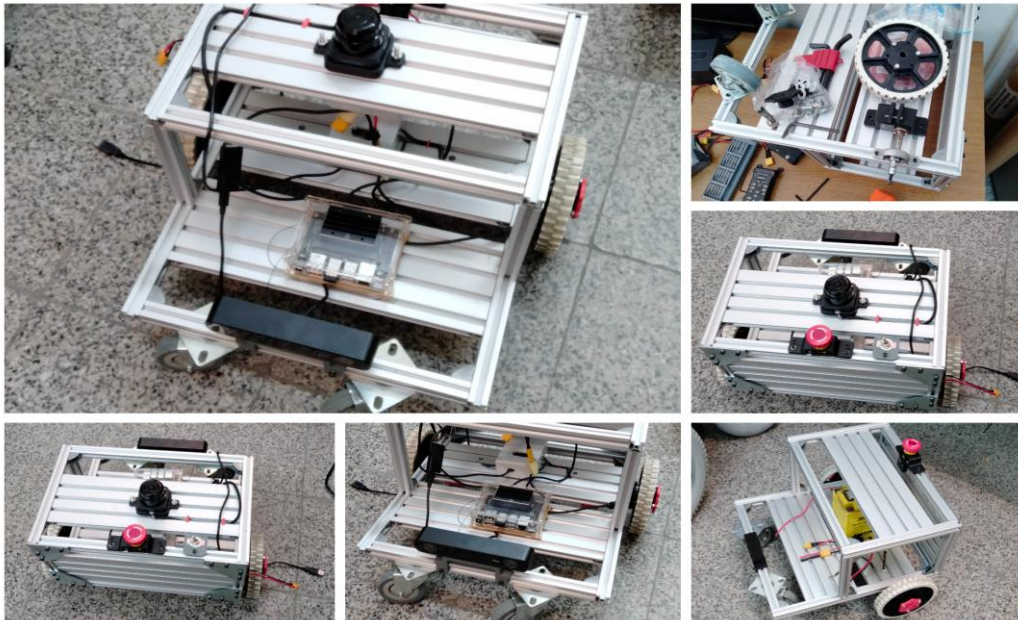


Hardware Design-The UGV



Hardware Design-The UGV

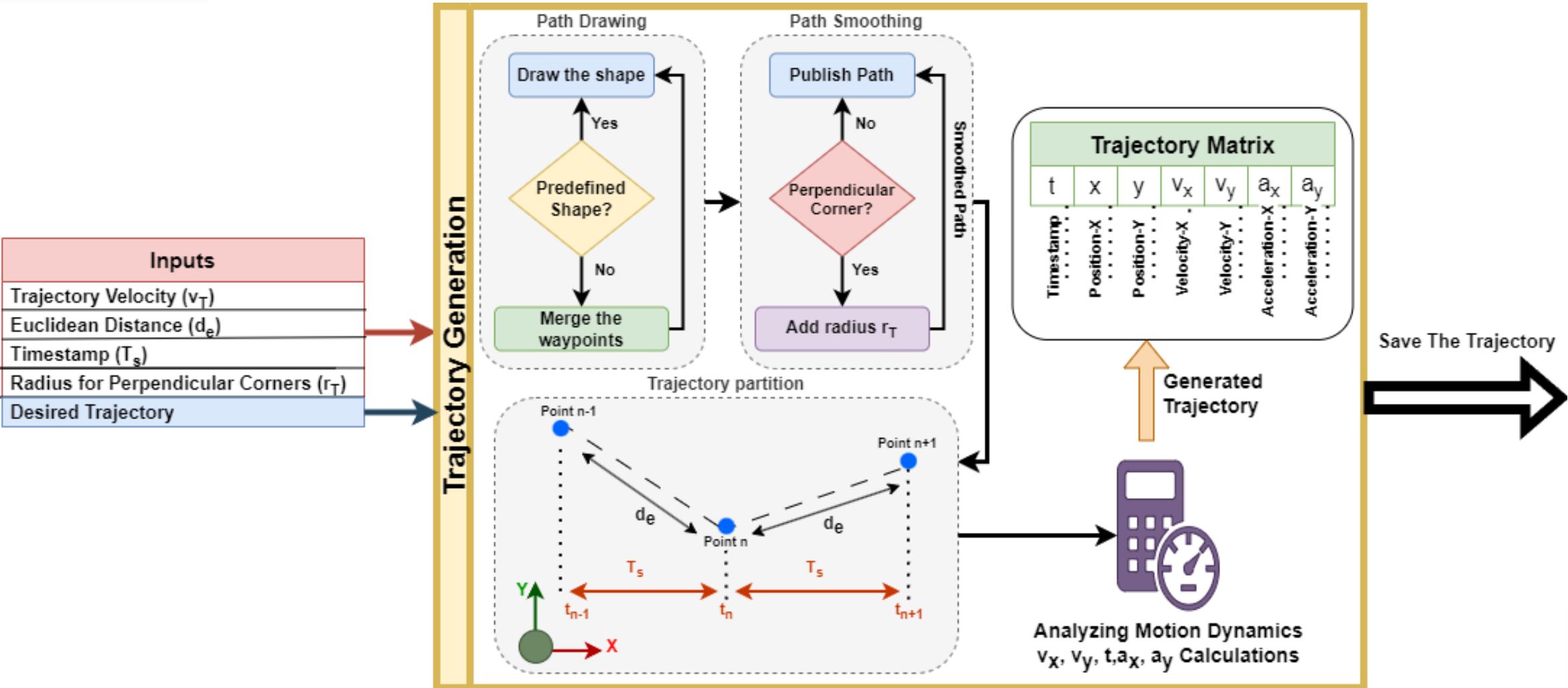
- Additionally, a small UGV was constructed for indoor testing using aluminum sigma profiles, and same sensor set.
- Initially, it was used to test algorithms in laboratory environments.



Navigation-Trajectory Generation

- Trajectory generation, vital for robotics motion planning, produces feasible paths considering various factors and avoiding obstacles.
- Its importance in this thesis is for localization comparison.
- The created trajectory should be feasible for the robot to evaluate the localization algorithms.
- The robot's localization is recorded and compared, with reduced discrepancies signaling improved localization.

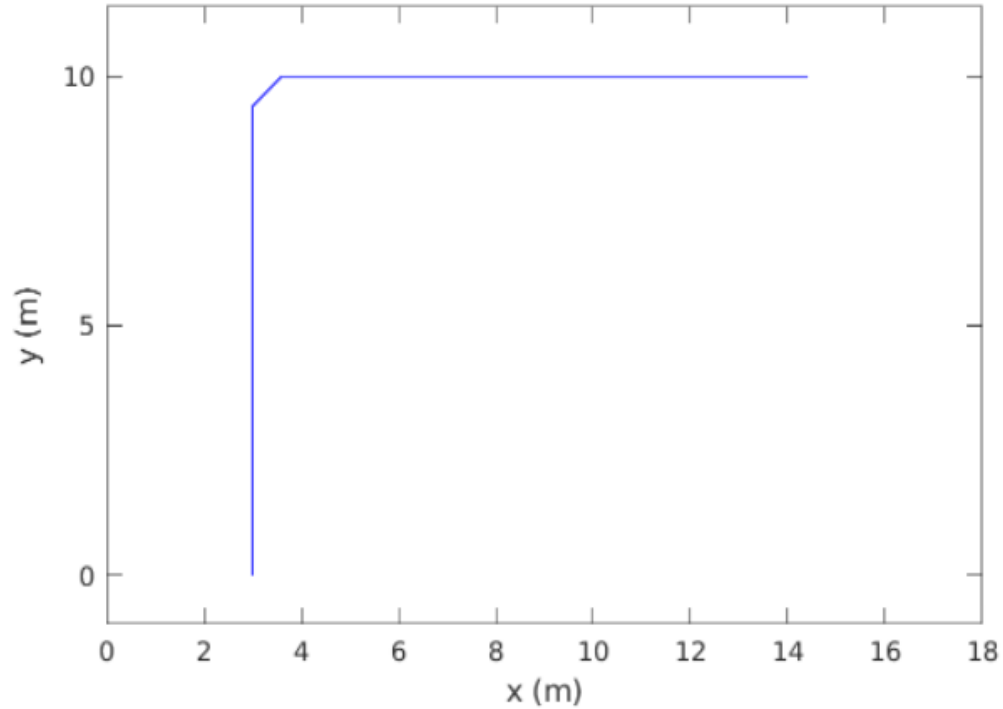
Navigation-Trajectory Generation



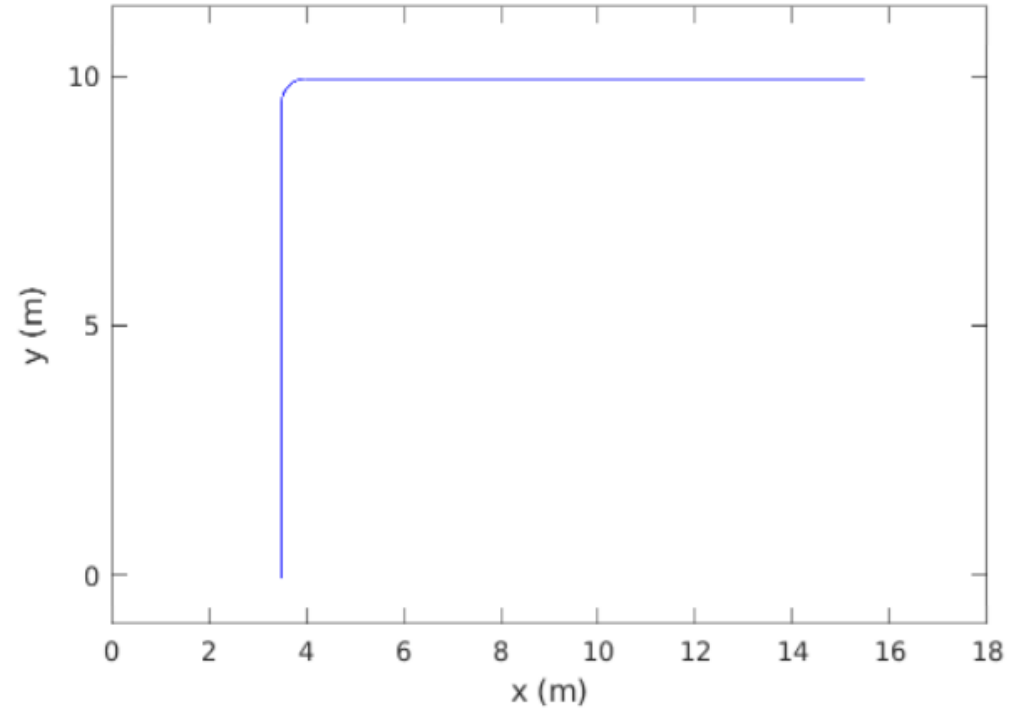
Navigation-Trajectory Generation

Path Smoothing

2D view



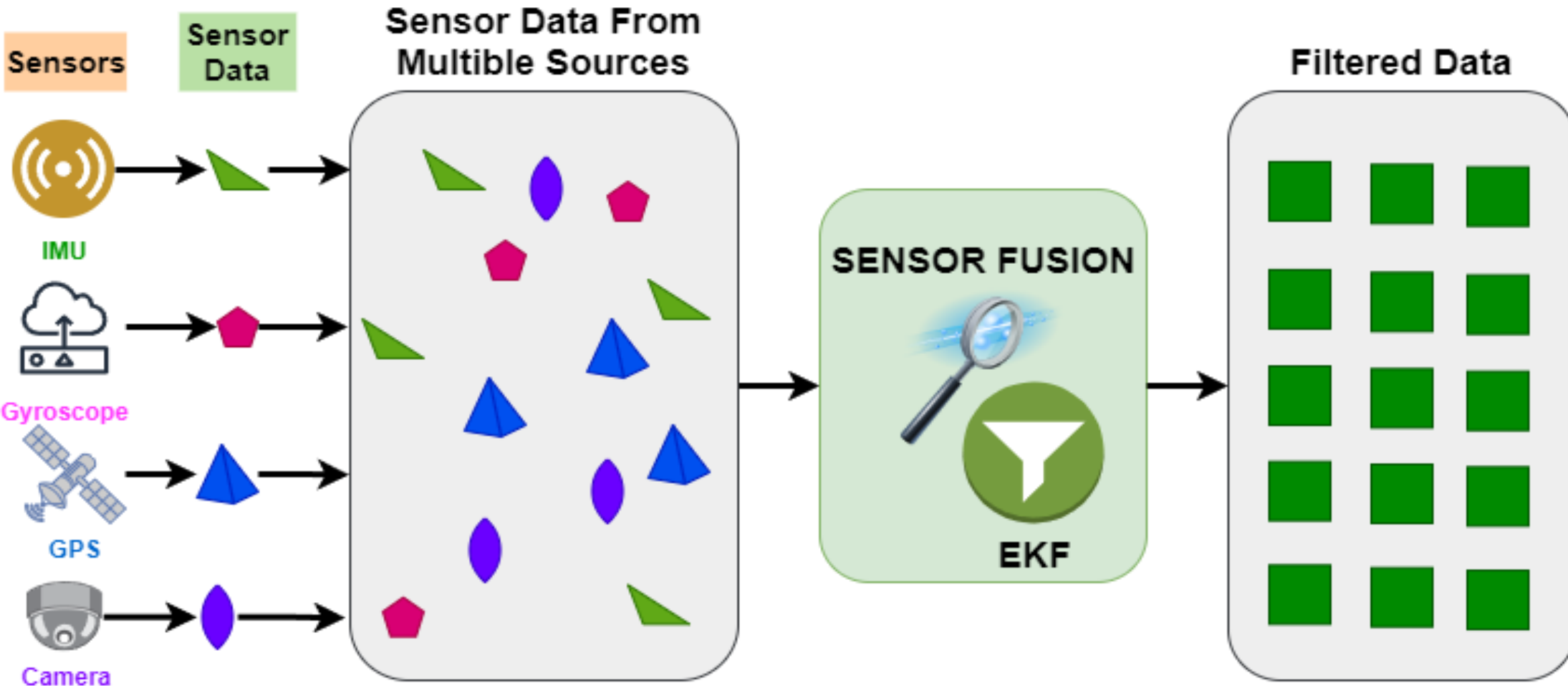
2D view



Navigation-Sensor Fusion

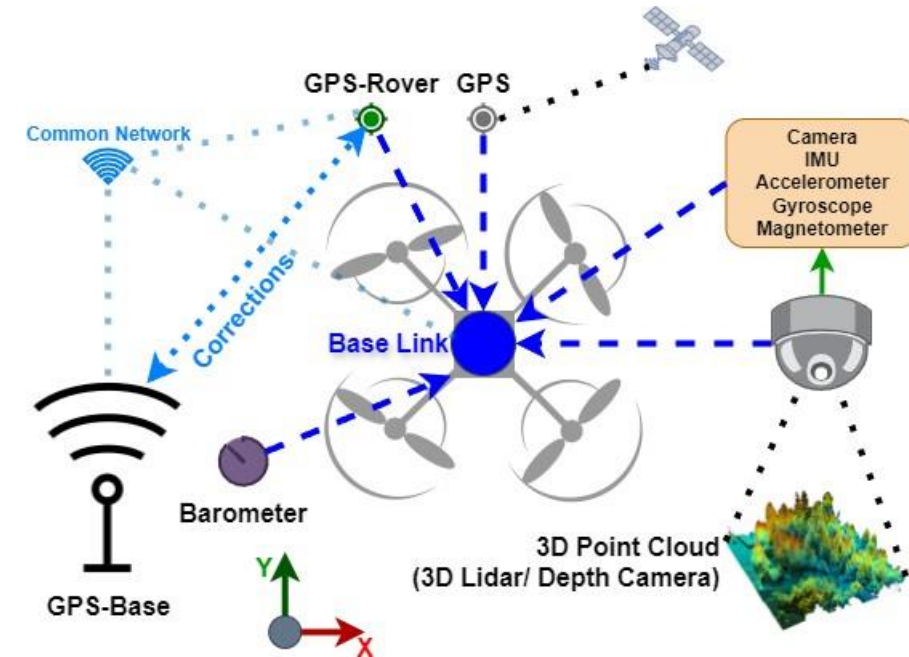
- Sensor fusion combines data from GPS, IMU, camera odometry, and gyroscope, enhancing robot accuracy.
- It improves localization performance by overcoming sensor limitations.
- Techniques include Bayesian filters, Kalman filters, and particle filters.
- This thesis applies the EKF technique, which consists of prediction and correction steps.

Navigation-Sensor Fusion



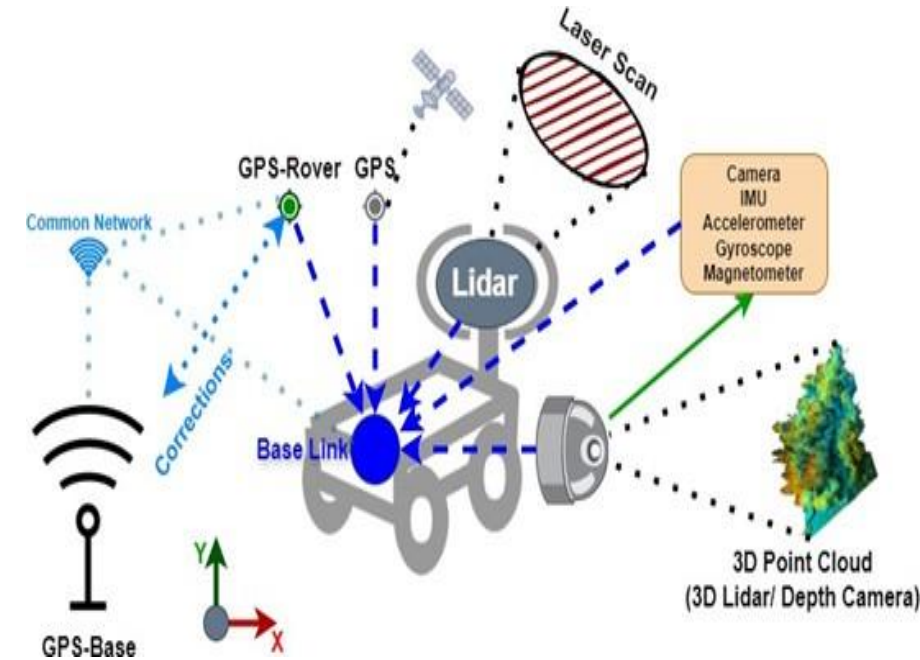
Navigation-Sensor Set for The UAV

Sensor Type	Differential	Source
IMU	Yes	ZEDm Stereo Camera
Gyroscope	Yes	ZEDm Stereo Camera
Accelerometer	Yes	ZEDm Stereo Camera
Stereo Camera	No	ZEDm Stereo Camera
Depth Camera (3D Lidar)	Yes	ZEDm Stereo Camera
GPS (Base Corrected)	No	Emlid Reach M+ RTK GNSS
GPS	No	Pixhawk Flight Controller
Barometer	No	Pixhawk Flight Controller



Navigation-Sensor Set for The UGV

Sensor Type	Differential	Source
IMU	Yes	ZED2 Stereo Camera
Gyroscope	Yes	ZED2 Stereo Camera
Accelerometer	Yes	ZED2 Stereo Camera
Stereo Camera	No	ZED2 Stereo Camera
Magnetometer	Yes	ZED2 Stereo Camera
Depth Camera (3D Lidar)	Yes	ZED2 Stereo Camera
GPS (Base Corrected)	No	Emlid Reach M+ RTK GNSS
2D Lidar	No	RPlidar S1



ROS & EKF Settings

UAV

Source Topic	Source Sensor(s)	Differential
\zedm\odom	ZEDm (Camera & 3D Lidar & Accelerometer)	Yes
\zedm\imu	ZEDm (IMU)	Yes
\zedm\pose	ZEDm (Camera & Gyroscope)	No
\tcp\fix	Emlid Reach M+ RTK GNSS (GPS)	No
\mavros\gps	Pixhawk Flight Controller (GPS)	No

UGV

Source Topic	Source Sensor(s)	Differential
\zed2\odom	ZED2 (Camera & 3D Lidar & Accelerometer)	Yes
\zed2\imu	ZED2 (IMU)	Yes
\zed2\pose	ZED2 (Camera & Gyroscope)	No
\tcp\fix	Emlid Reach M+ RTK GNSS (GPS)	No

Value/Content	Matrix Elements
Position	$x \mid y \mid z$
Orientation	$\phi \mid \theta \mid \psi$
Linear Velocity	$\dot{x} \mid \dot{y} \mid \dot{z}$
Angular Velocity	$\dot{\phi} \mid \dot{\theta} \mid \dot{\psi}$
Linear Acceleration	$\ddot{x} \mid \ddot{y} \mid \ddot{z}$

Element	Odom	Pose	IMU
x	True	True	False
y	True	True	False
z	True	True	False
ϕ	False	False	False
θ	False	False	False
ψ	True	True	False
$\dot{x}$	True	False	True
$\dot{y}$	True	False	True
$\dot{z}$	True	False	True
$\dot{\phi}$	False	False	True
$\dot{\theta}$	False	False	True
$\dot{\psi}$	True	False	True
$\ddot{x}$	True	False	True
$\ddot{y}$	True	False	True
$\ddot{z}$	True	False	True

Navigation-Control Theory

- ♦ Proportional-Integral-Derivative (PID) Control:
- ♦ The PID controller is a straightforward control law.
- ♦ It considers the error variable (P for "proportional"), the integral (I for "integral"), and the derivative of the error variable (D for "derivative").
- ♦ The time-domain control law is represented by the equation below.

$$U(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

Navigation-Control Theory

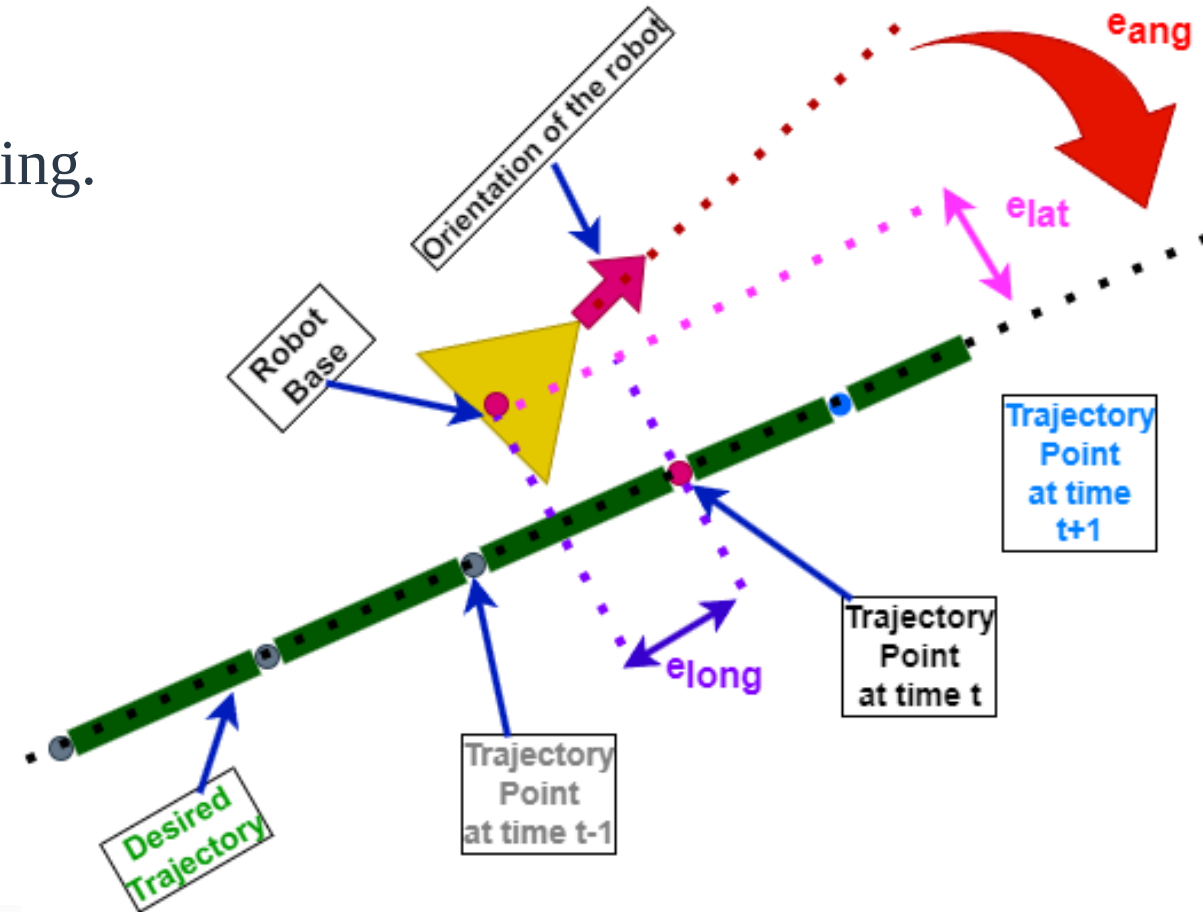
There are 3 PID controllers for trajectory following.

Longitudinal

Lateral

Angular

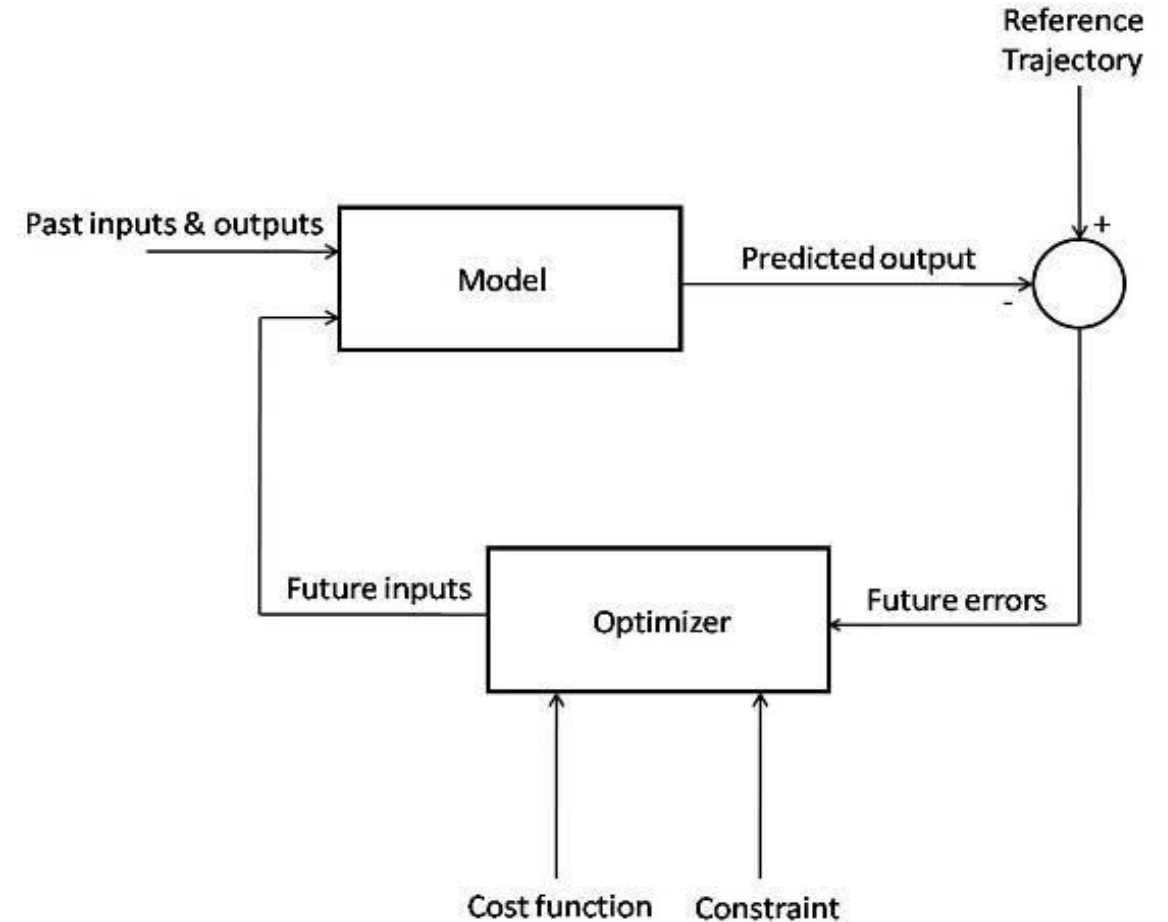
ROS Package: Tracking_PID
(Modified)



Navigation-Control Theory

Model Predictive Control:

- MPC is a more advanced control concept compared to the PID controller.
- In MPC, an internal model is used at each time step to predict system behavior over a predefined horizon.
- An optimal control input sequence that minimizes a certain cost function is generated.



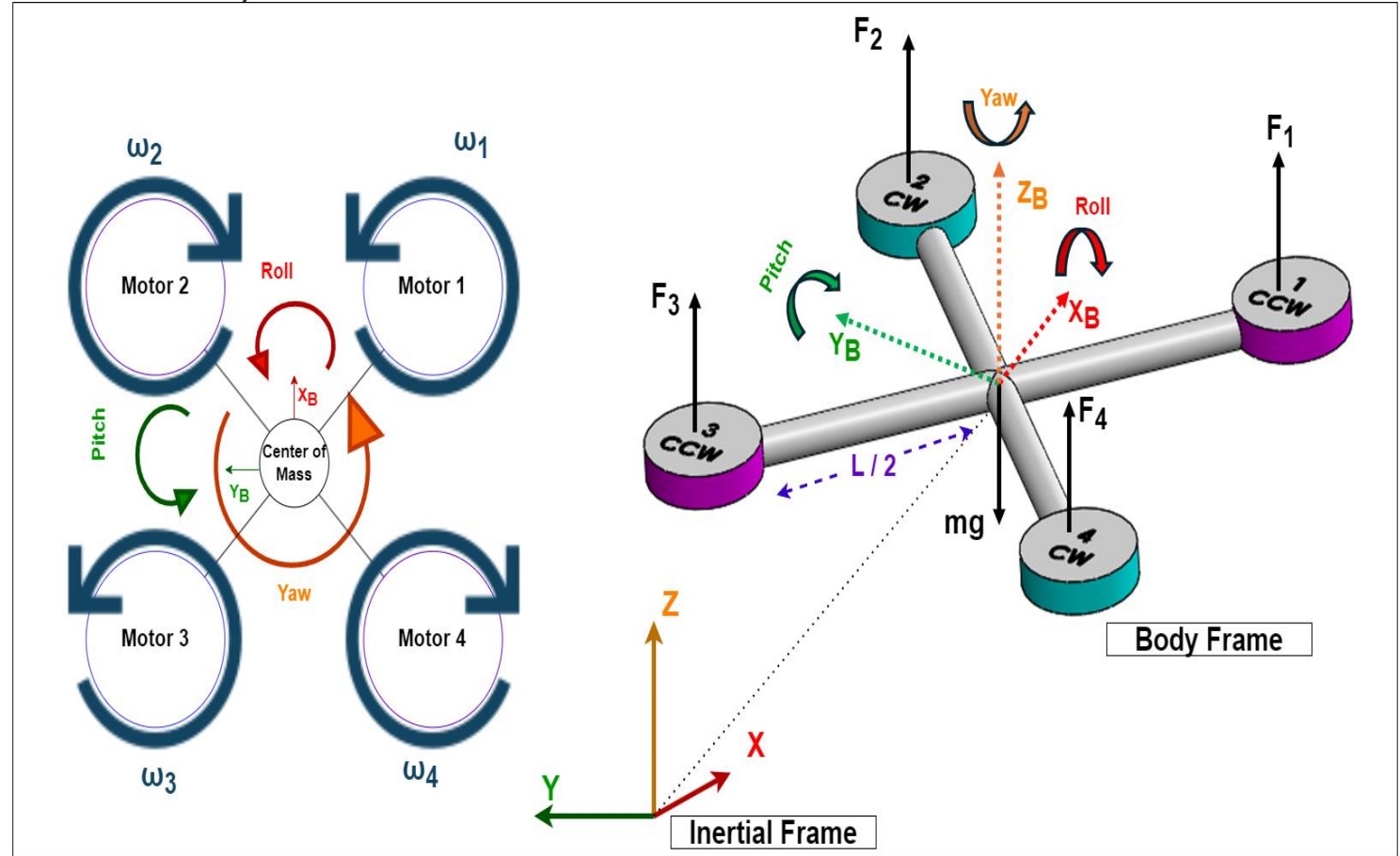
Kinematics and Dynamics Analysis of the UAV

The IF is the coordinate frame where the quadcopter moves. The quadcopter's position and orientation can be denoted as

$$\xi = [x \ y \ z]^T$$

$$\eta = [\phi \ \theta \ \psi]^T$$

where $\xi \in \mathbb{R}^3$ is the position and $\eta \in \mathbb{R}^3$ is the orientation in the IF.



Kinematics and Dynamics Analysis of the UGV

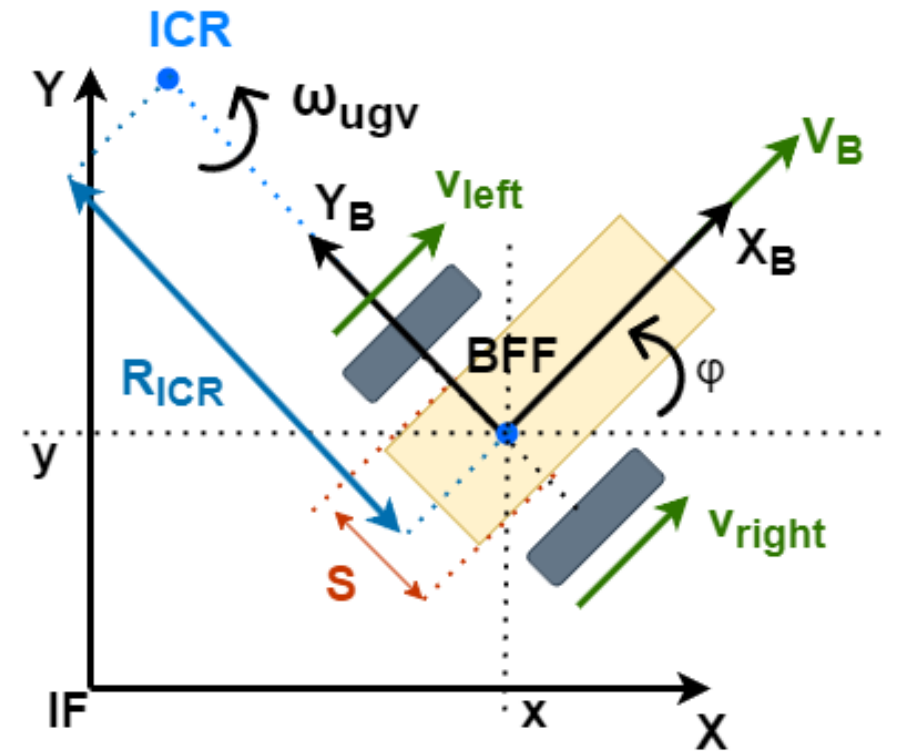
The translational velocities of both wheels and the angular velocity ω_{ugv} , and the translational velocity of the robot in the BFF can be calculated using equations below

$$v_{left} = \omega_{ugv}(R_{ICR} - S/2)$$

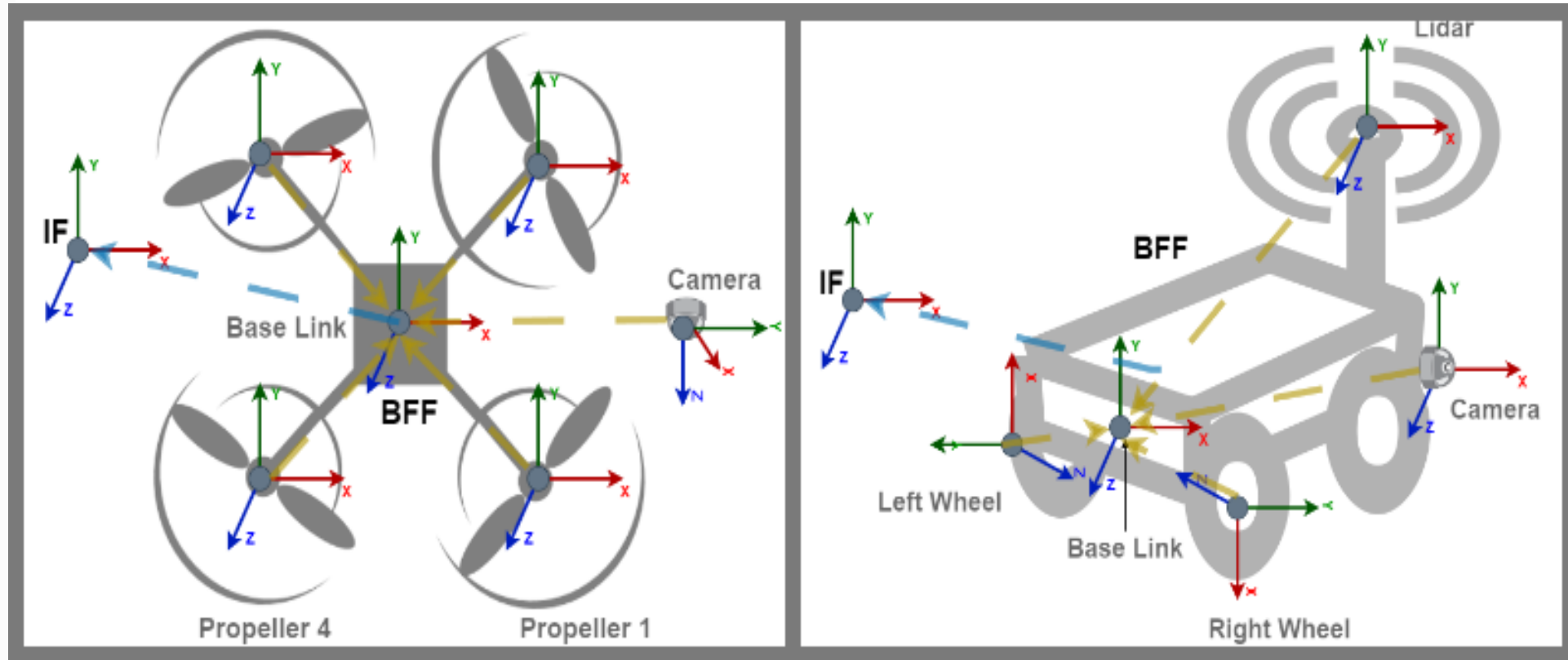
$$v_{right} = \omega_{ugv}(R_{ICR} + S/2)$$

$$\omega_{ugv} = \frac{v_{right} - v_{left}}{S}$$

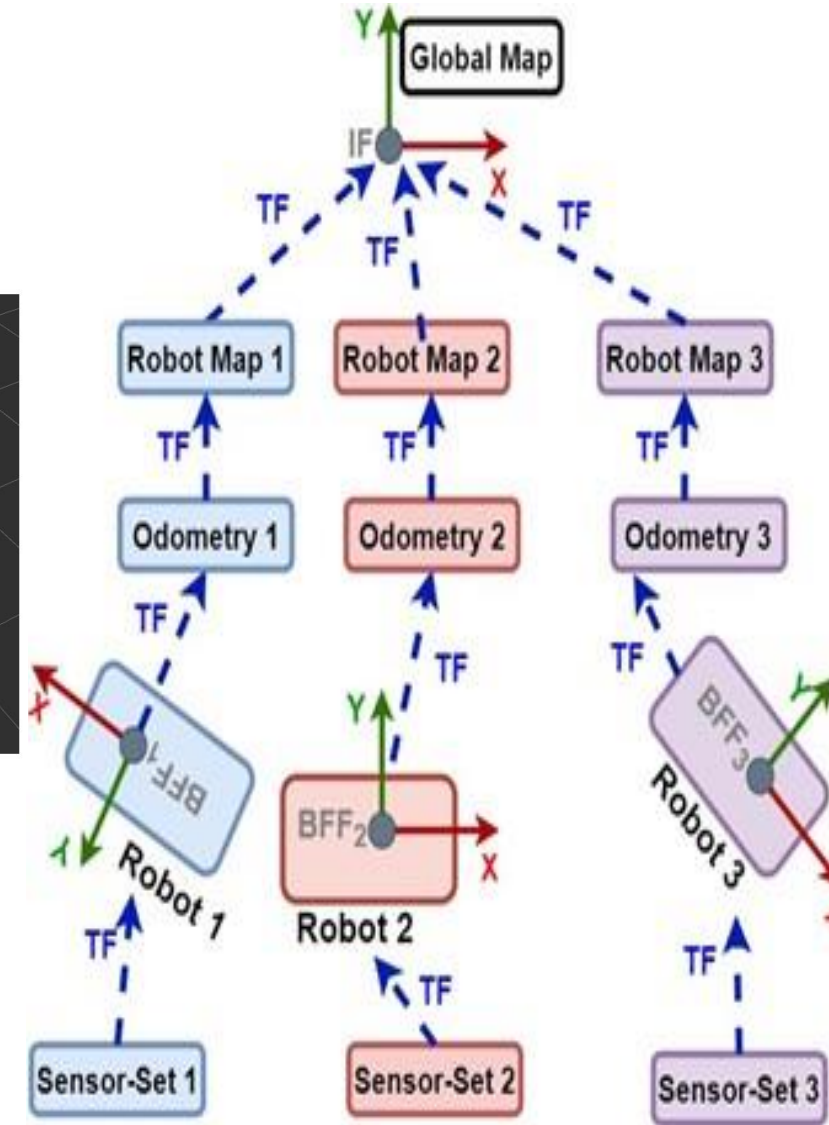
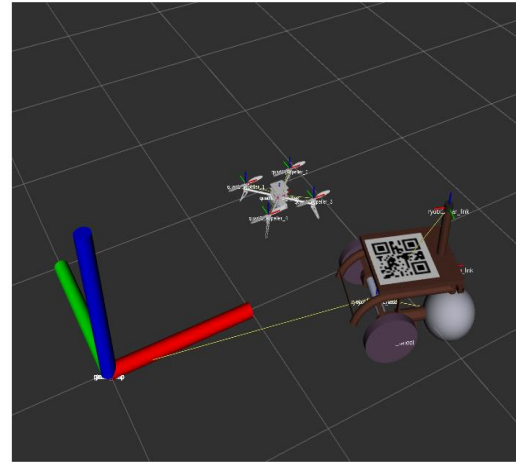
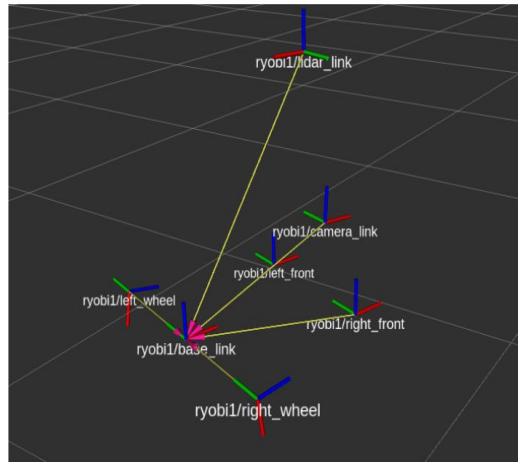
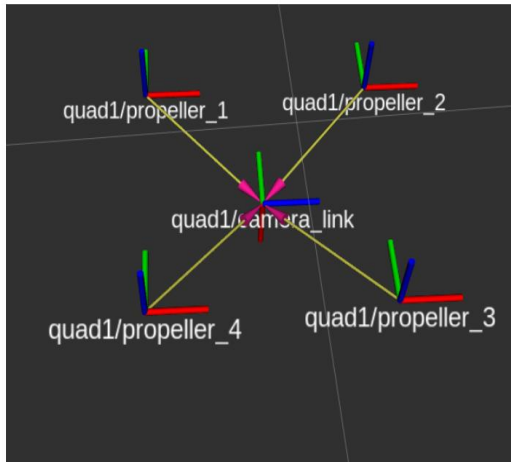
$$v_{ugv} = \frac{v_{right} + v_{left}}{2}$$



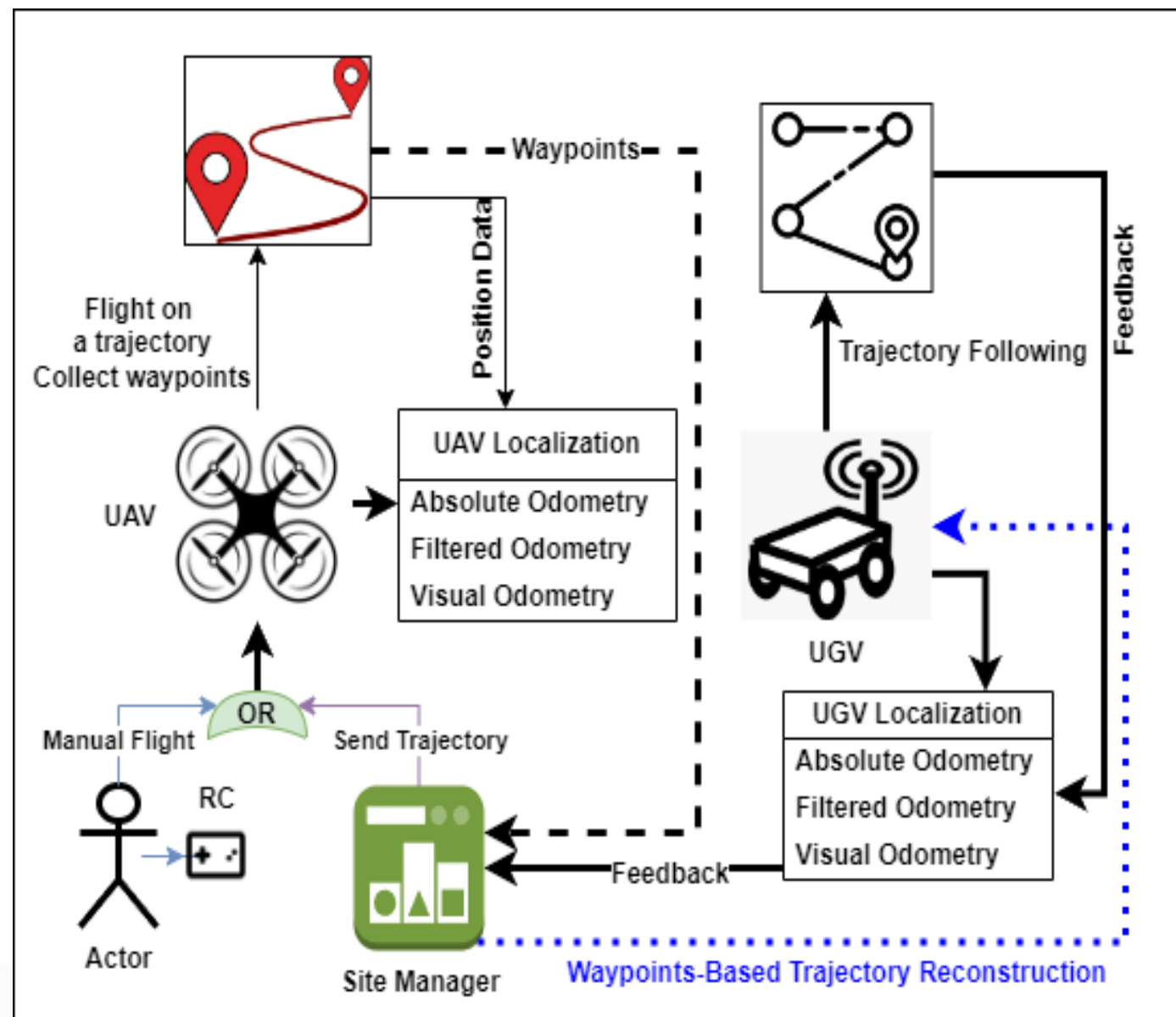
Coordinate Transformations



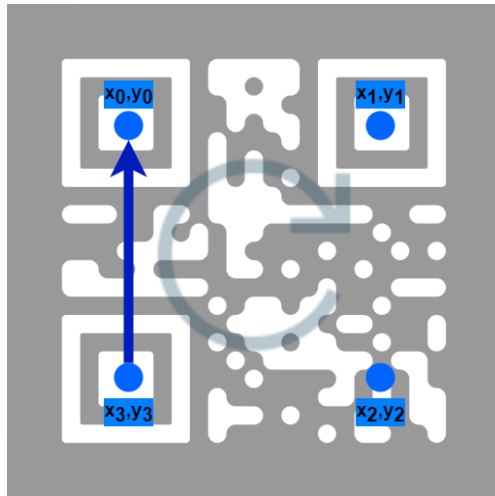
Coordinate Transformations



Coordinate Mission 1



Coordinate Mission 2

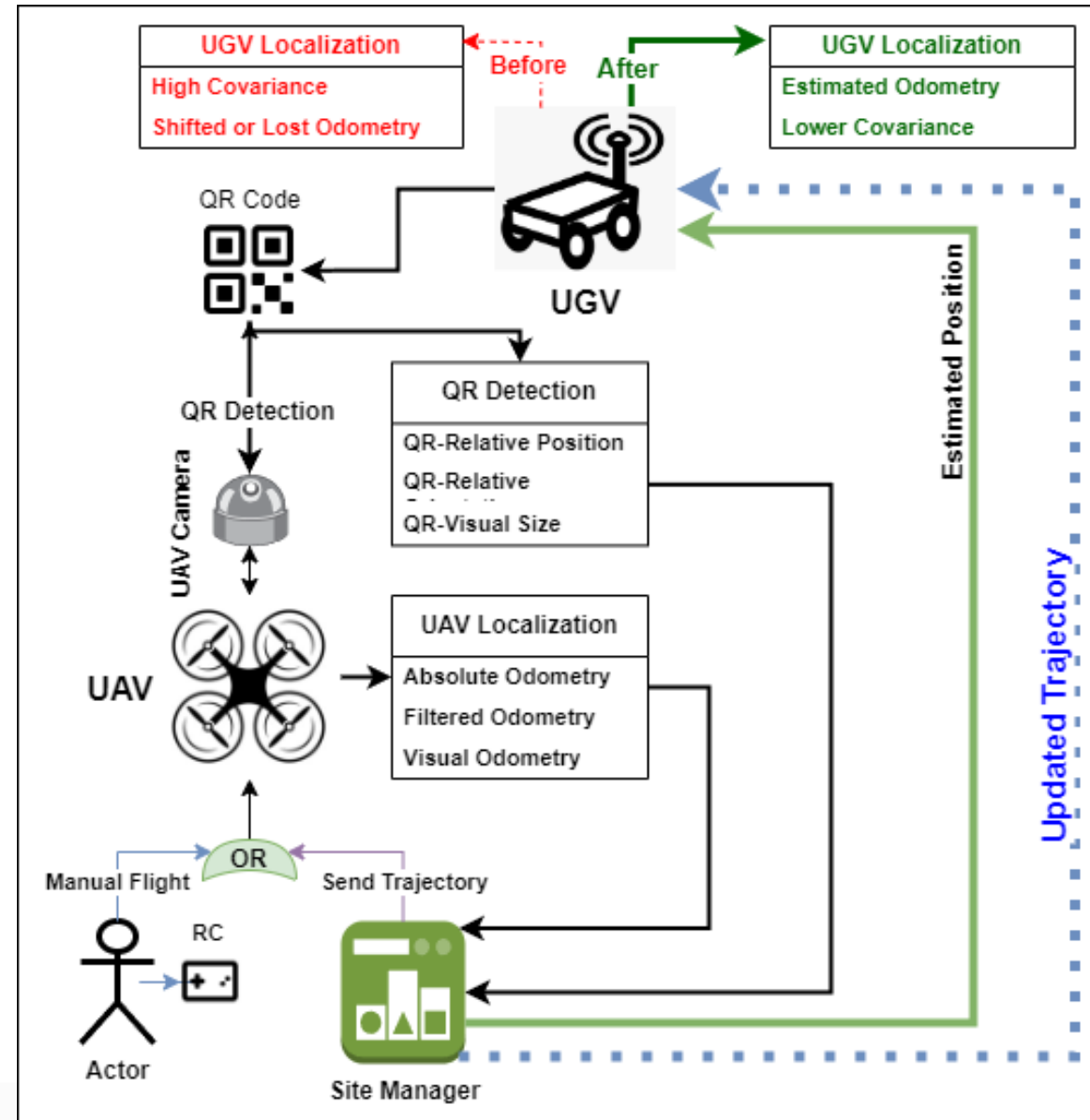


$$\text{vector_x} = x_3 - x_0$$

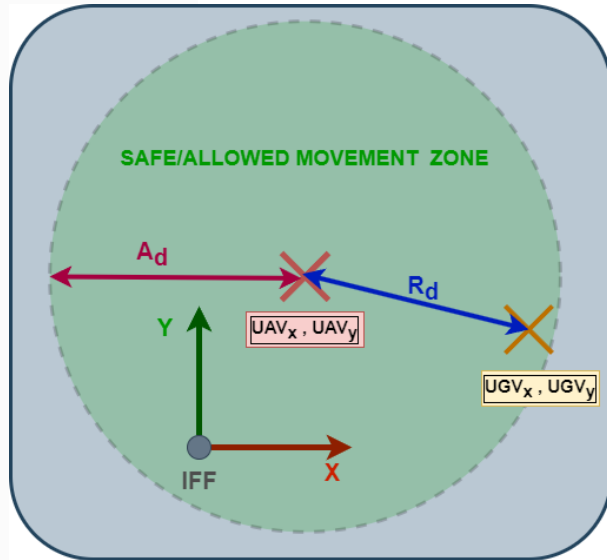
$$\text{vector_y} = y_3 - y_0$$

$$\text{QR-Code's orientation-radians} = \arctan\left(\frac{\text{vector}_y}{\text{vector}_x}\right)$$

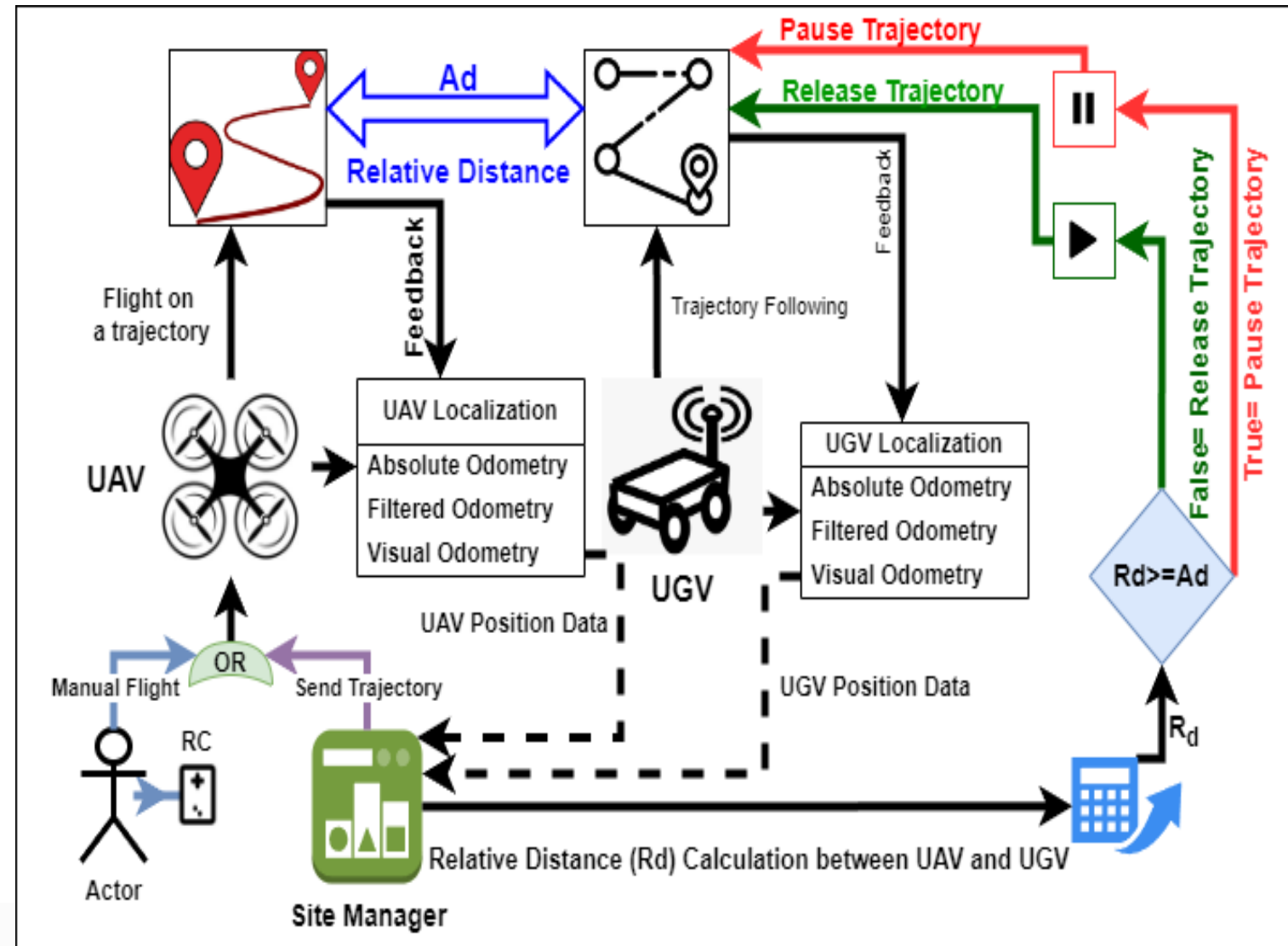
$$\text{QR-Code's orientation-degree} = \text{QR-Code's orientation-radians} \times \left(\frac{180}{\pi}\right)$$



Coordinate Mission 3



$$R_d = \sqrt{(UAV_x - UGV_x)^2 + (UAV_y - UGV_y)^2}$$

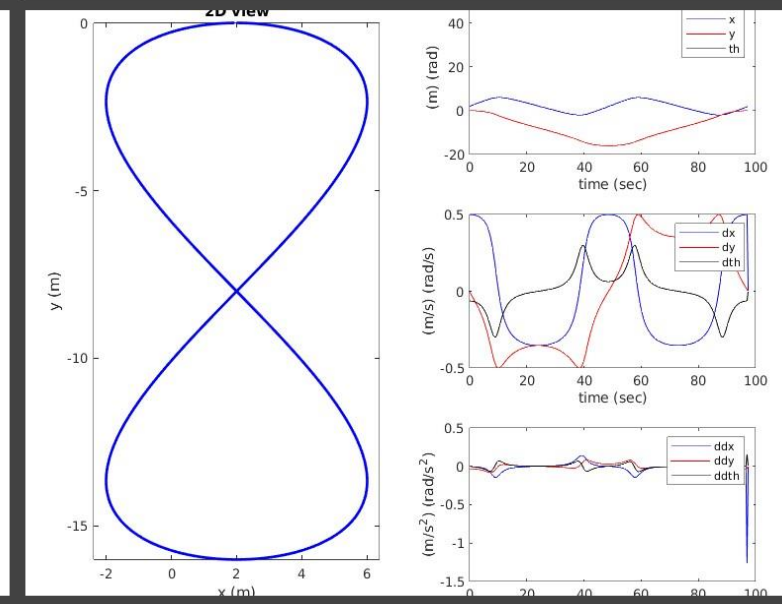
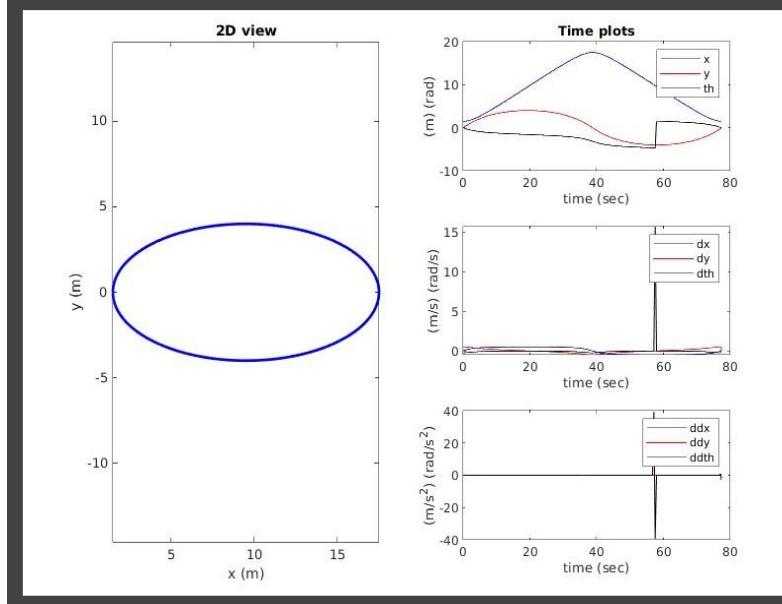
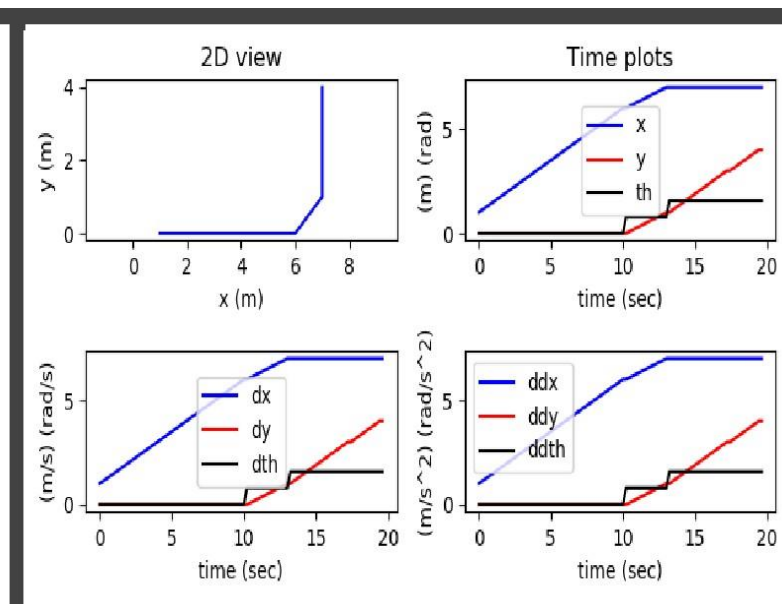
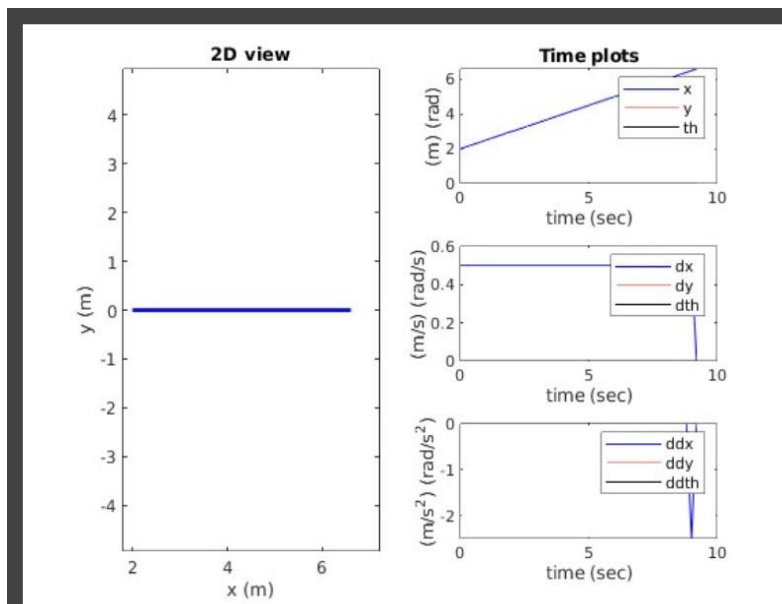


Simulation Works

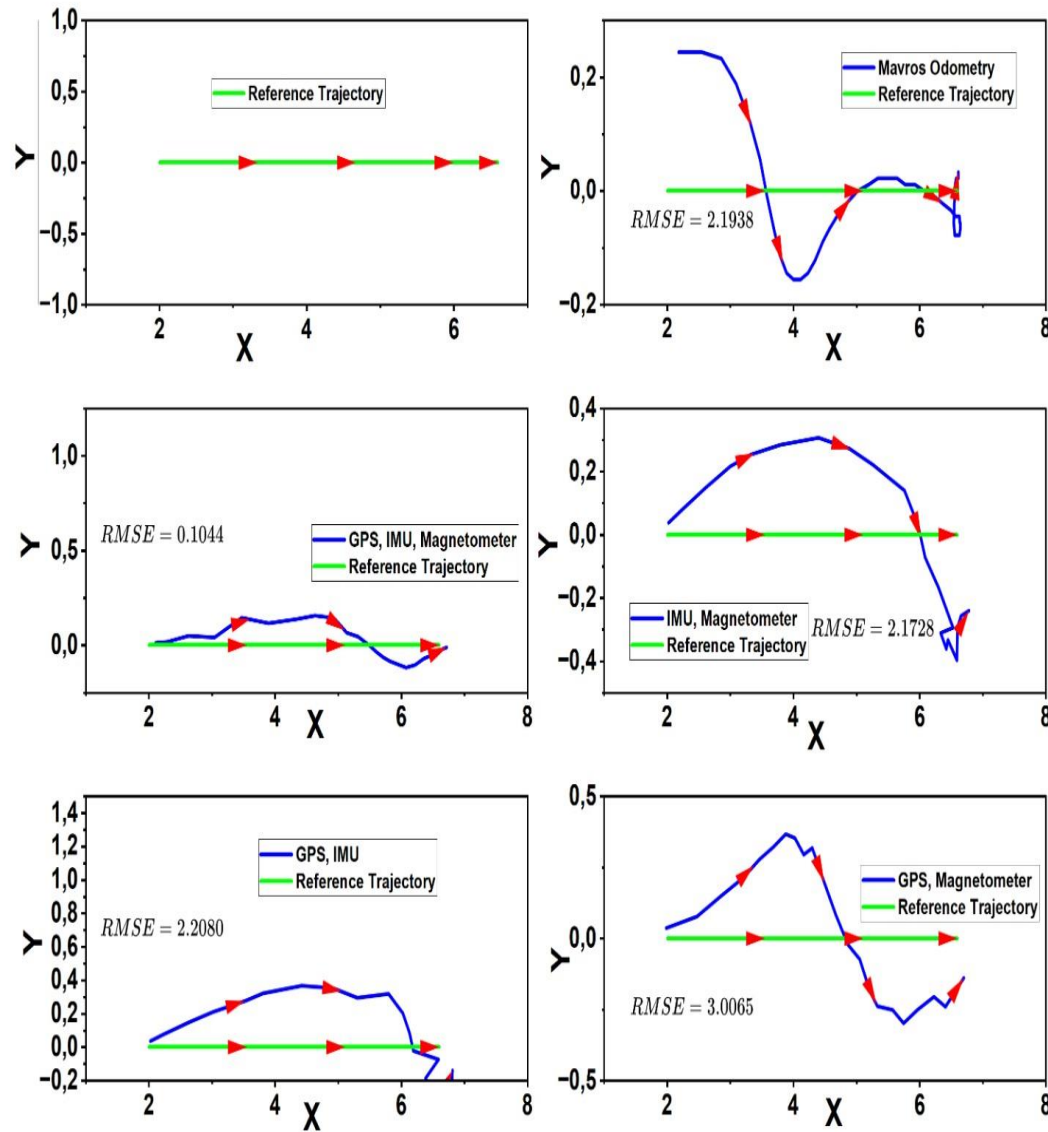
Parameter Name	Value	Unit
Velocity	0.5	meter/second
Time step	0.05	second
Euclidian distance	0.1	meter
Radius of corners	1	meter

Parameter	Longitudinal	Lateral	Angular
K_p	1.0	-3	1.5
K_i	0.1	-0.008	0.1
K_d	0.2	-1.0	0.05

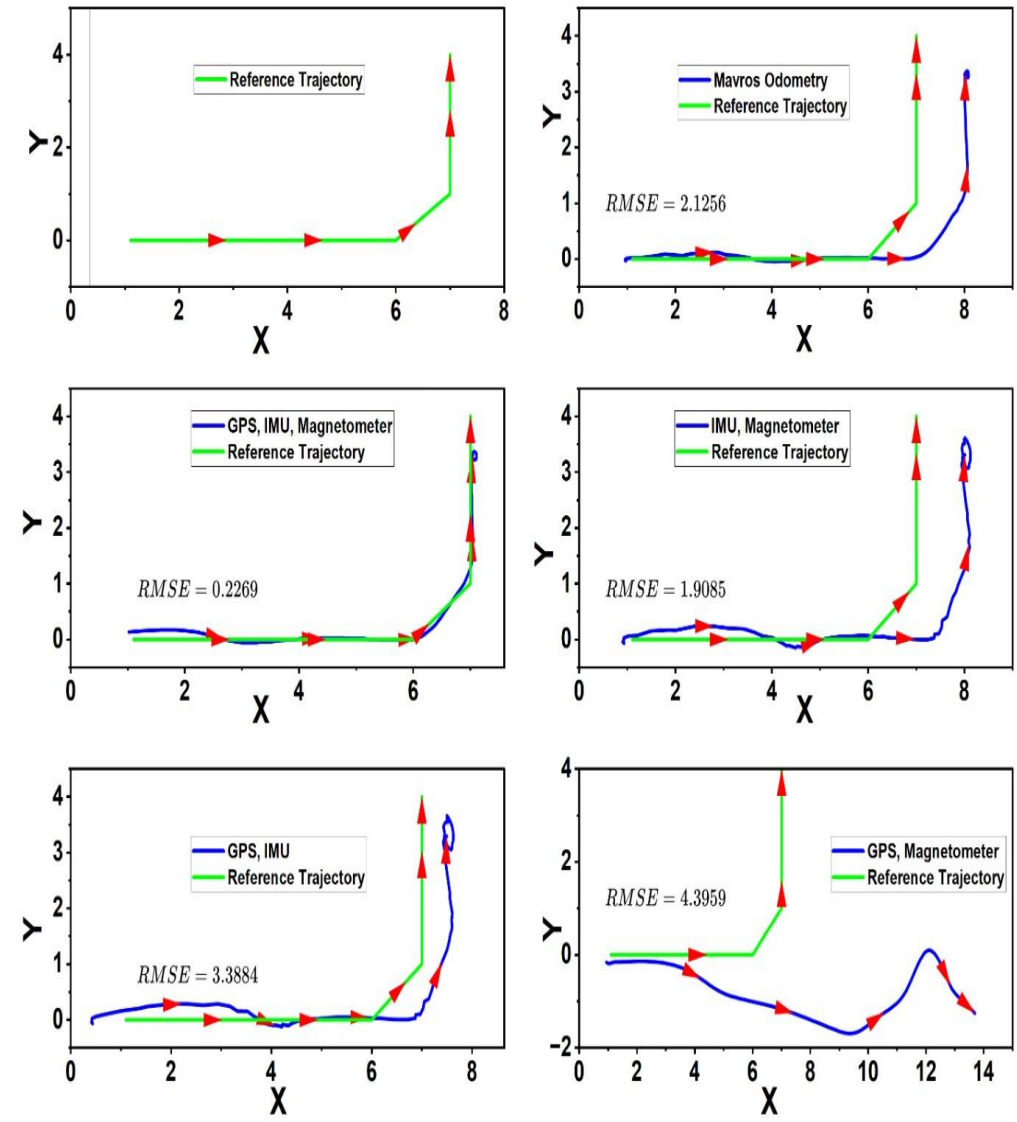
Number	Configuration
Config 1	Mavros Odometry-No EKF
Config 2	GPS, IMU and Magnetometer
Config 3	IMU, and Magnetometer
Config 4	GPS and IMU
Config 5	GPS and Magnetometer



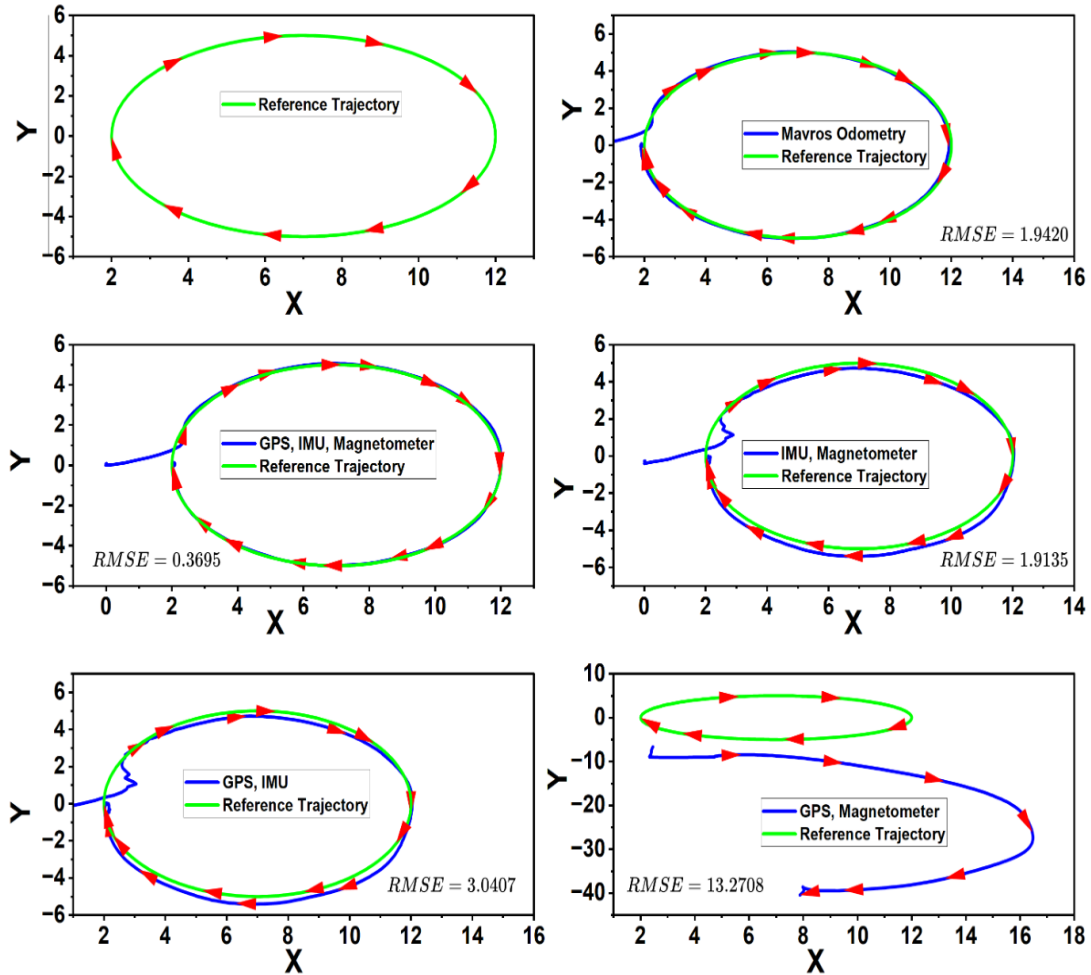
Localization Plots for The UAV: Line



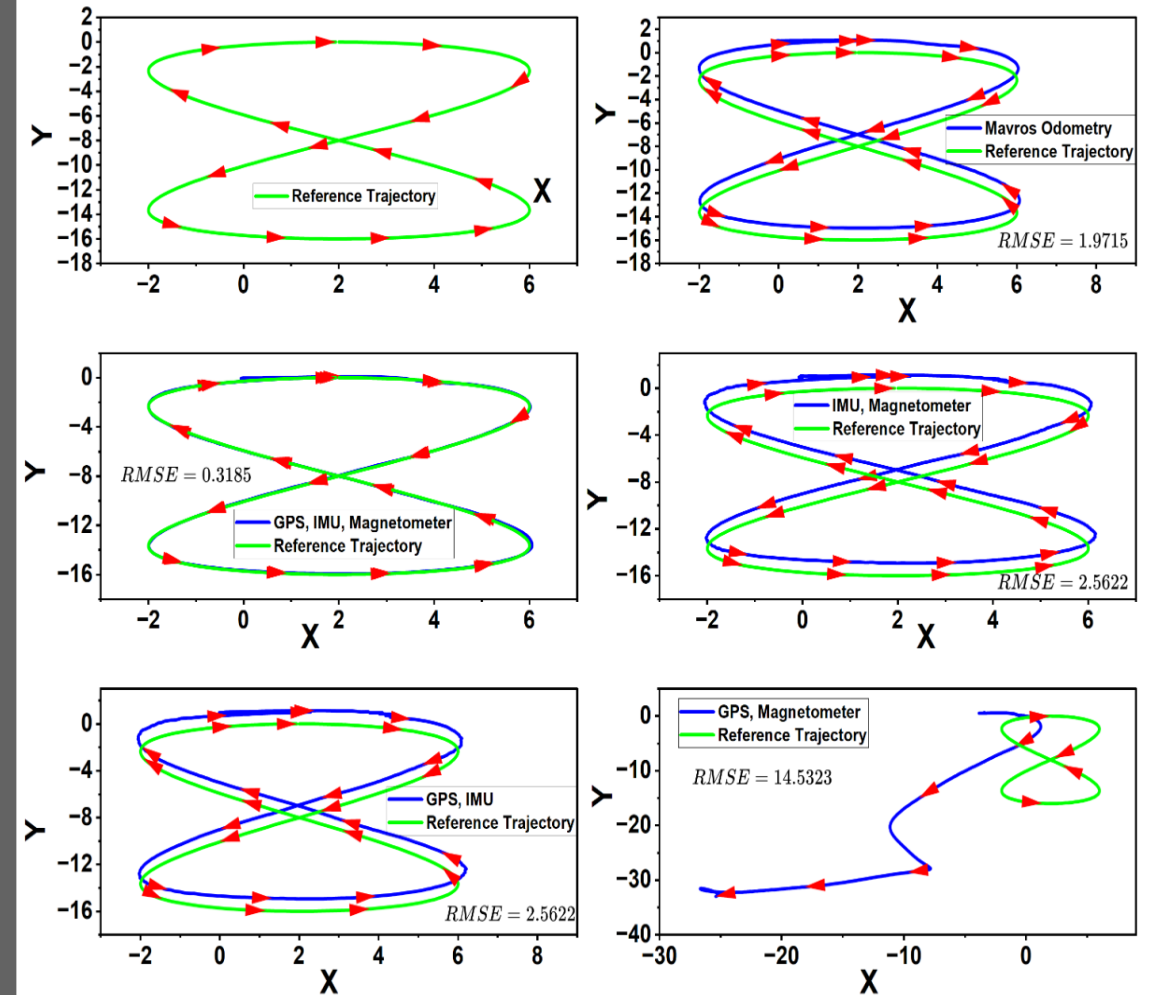
Localization Plots for The UAV: L-Shaped Figure

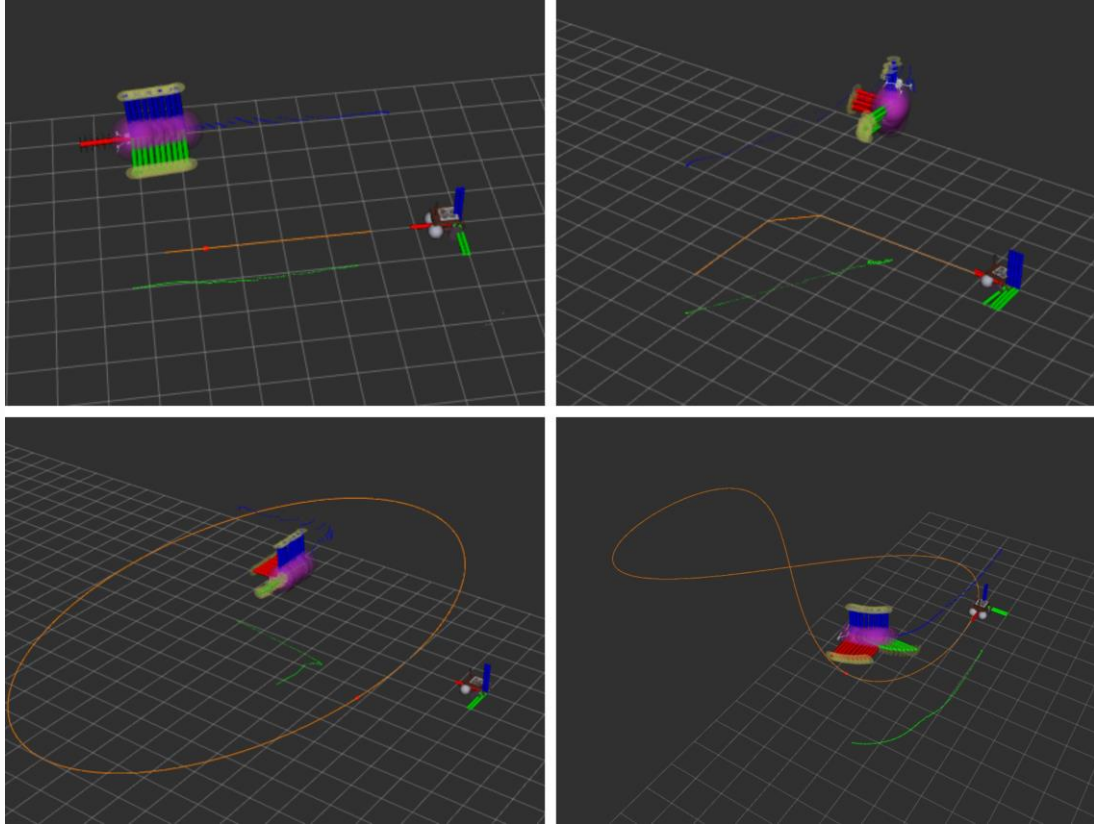


Localization Plots for The UAV: Ellipse



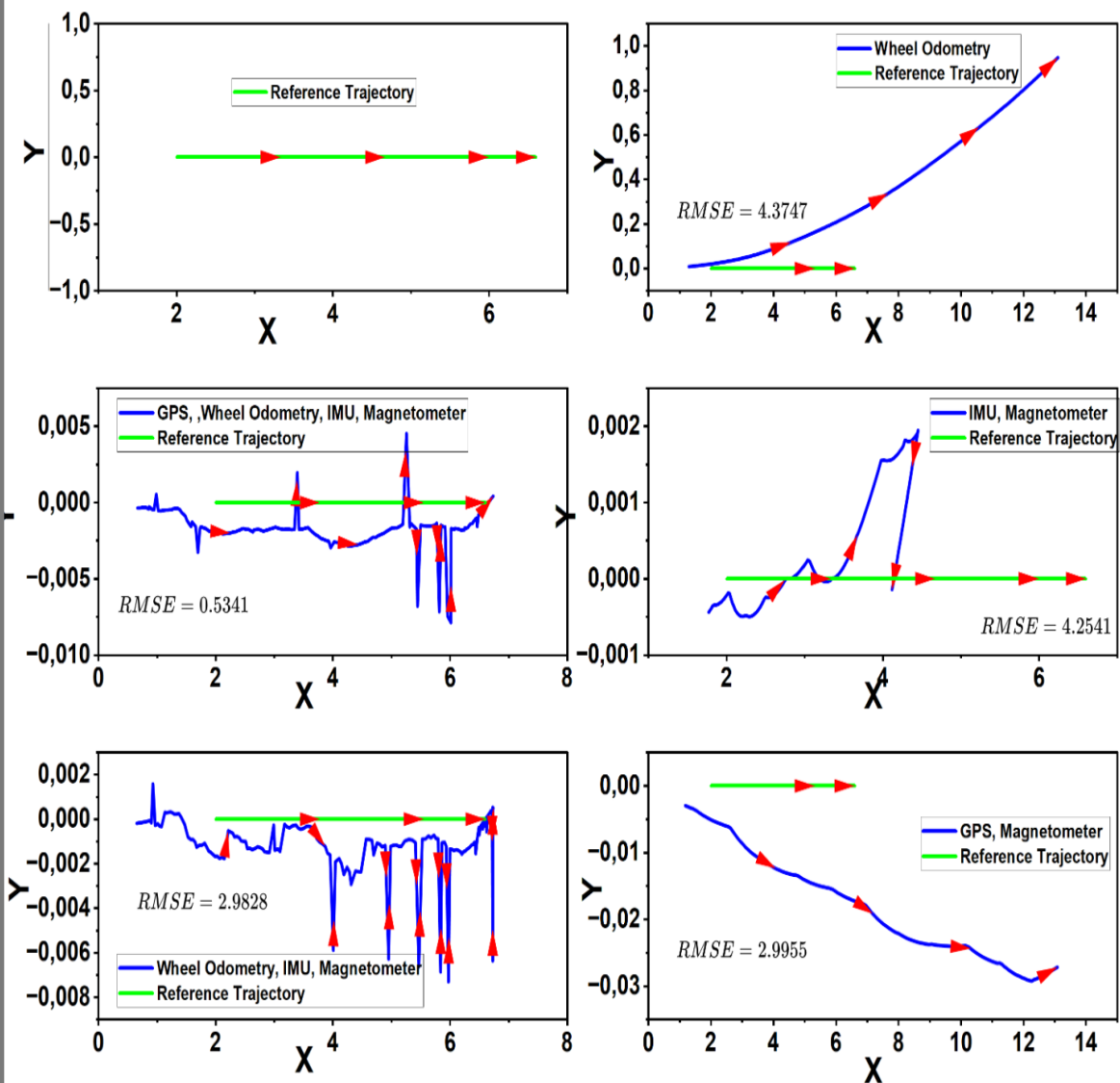
Localization Plots for The UAV: 8-Figure



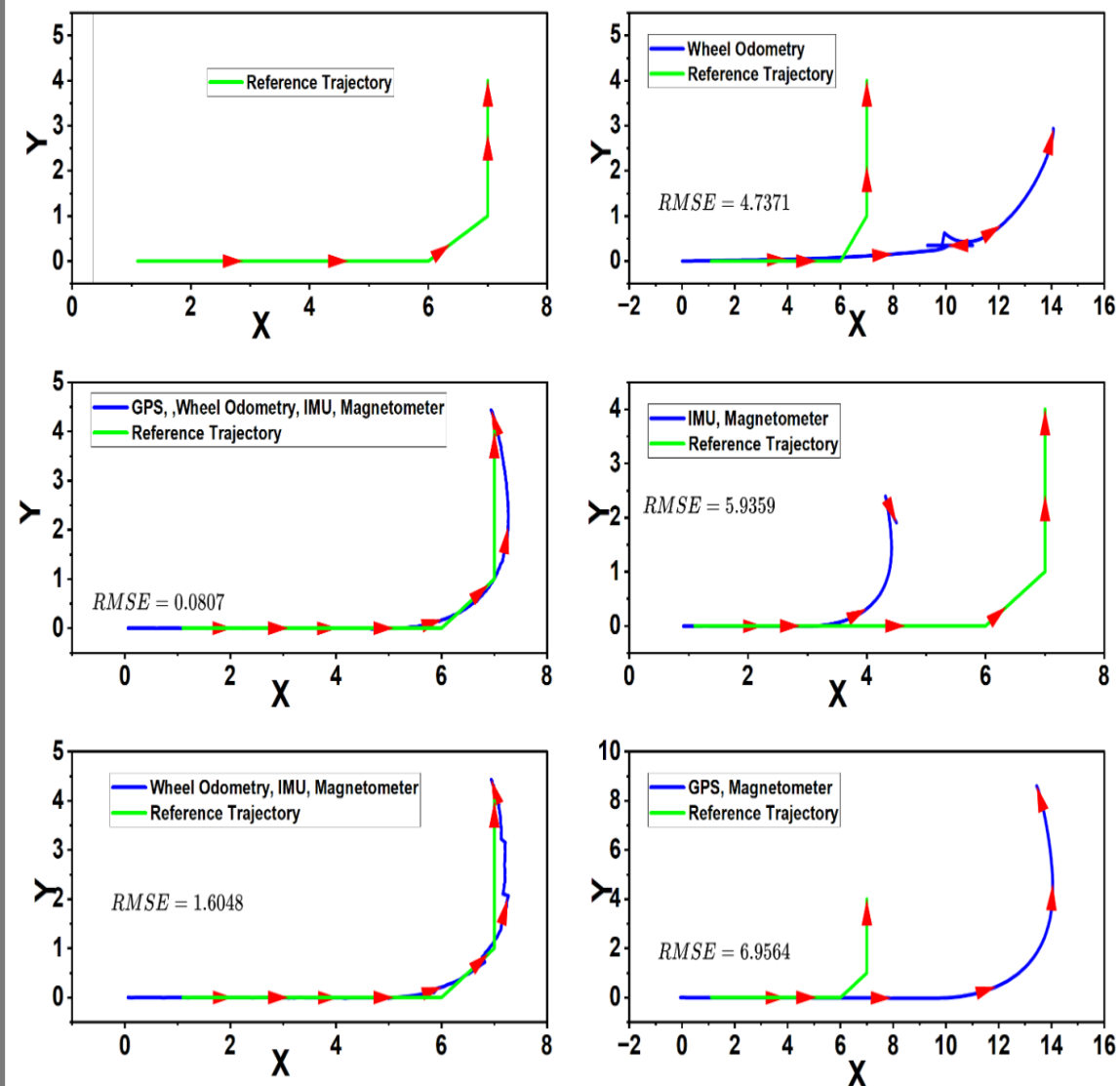


Trajectory	GPS	IMU	Magnetometer	Mavros Odometry	RMSE
Line	-	-	-	X	2,1938
	X	X	X	X	0,1044
	-	X	X	-	2,1728
	-	X	X	X	2,2080
	X	-	X	-	3,0065
L-Shaped Figure	-	-	-	X	2,1256
	X	X	X	X	0,2269
	-	X	X	-	1,9085
	-	X	X	X	3,3884
	X	-	X	-	4,3959
Ellipse	-	-	-	X	1,9420
	X	X	X	X	0,3695
	-	X	X	-	1,9135
	-	X	X	X	3,0407
	X	-	X	-	13,2708
8-Figure	-	-	-	X	1,9715
	X	X	X	X	0,3185
	-	X	X	-	2,5622
	-	X	X	X	2,5622
	X	-	X	-	14,5323

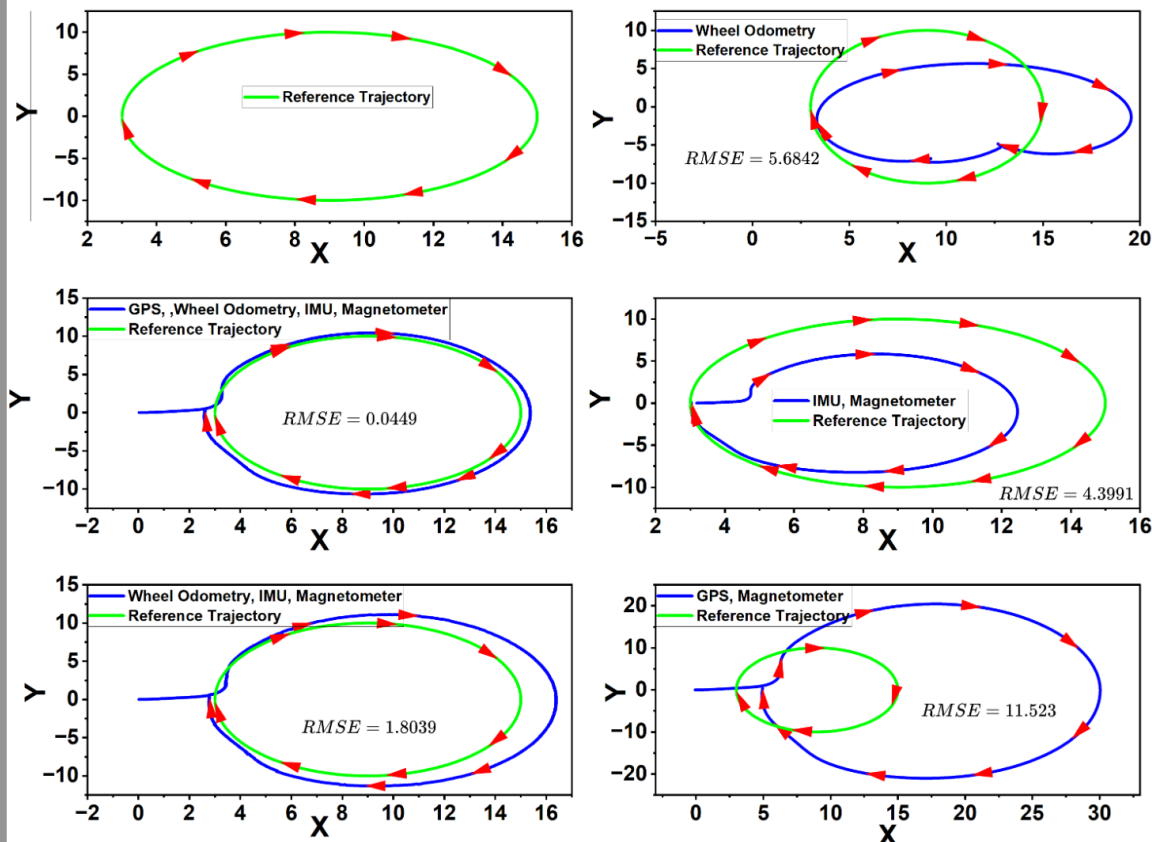
Localization Plots for The UGV: Line



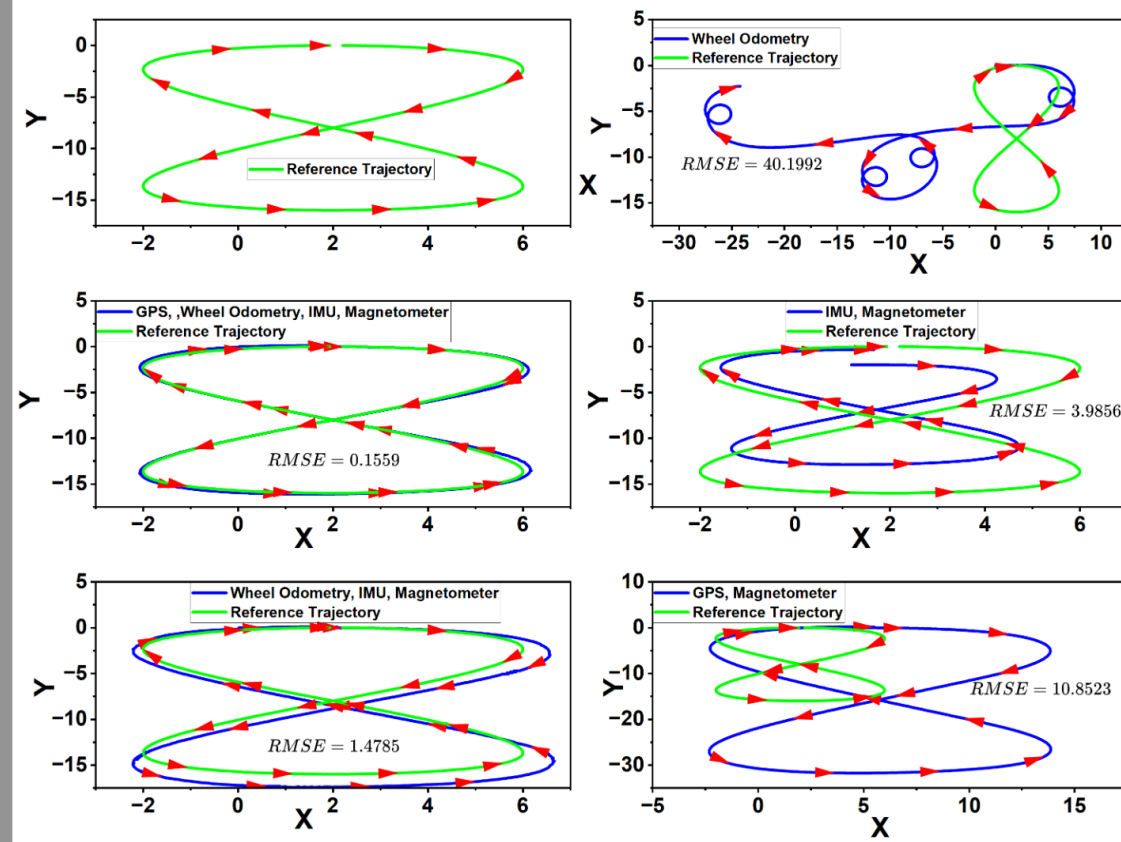
Localization Plots for The UGV: L-Shape



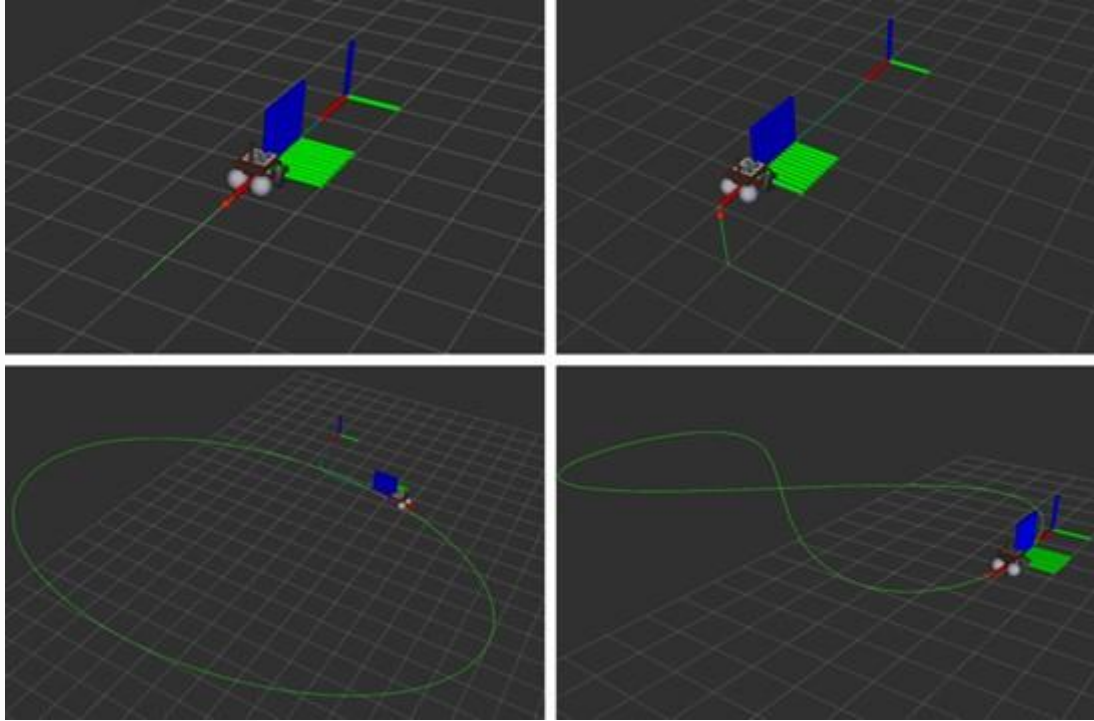
Localization Plots for The UGV: Ellipse



Localization Plots for The UGV: 8-Figure



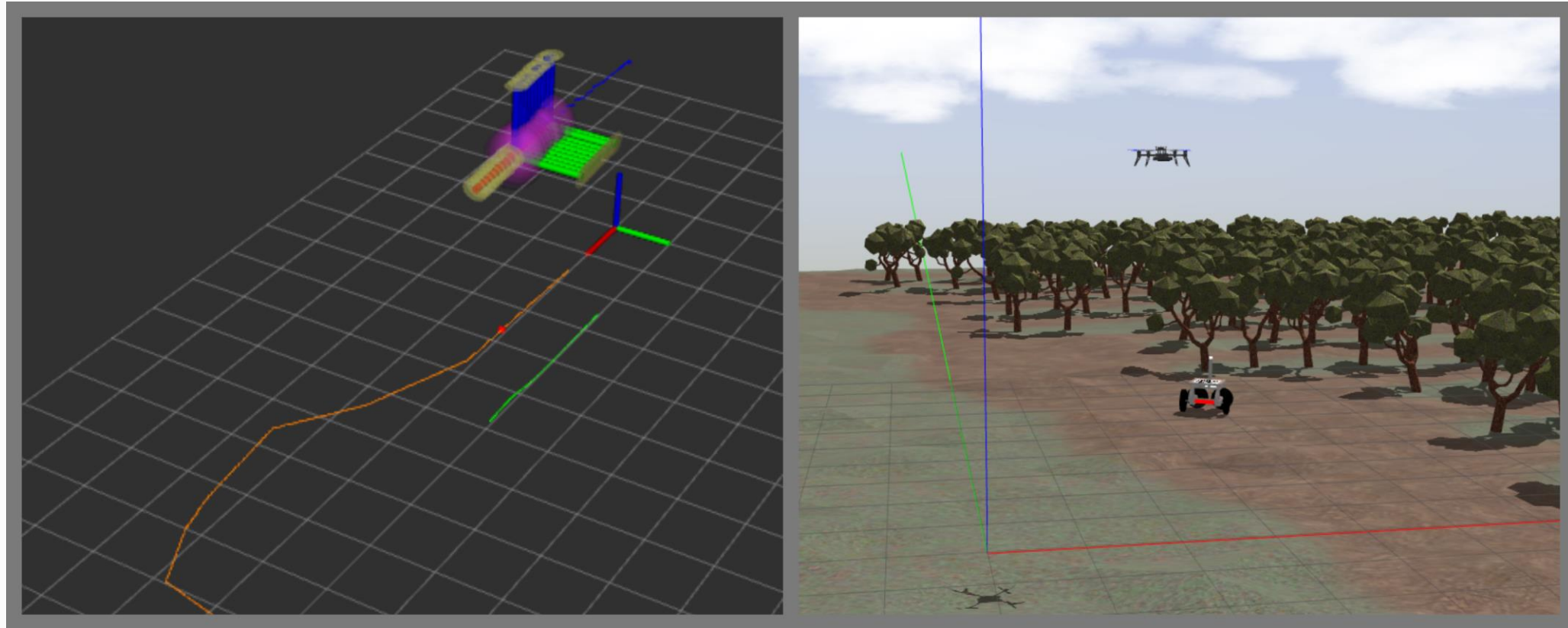
Parameter	Longitudinal	Lateral	Angular
K_p	0.3	-0.8	-0.05
K_i	0.01	-0.001	-0.001
K_d	0.2	-0.01	-0.01



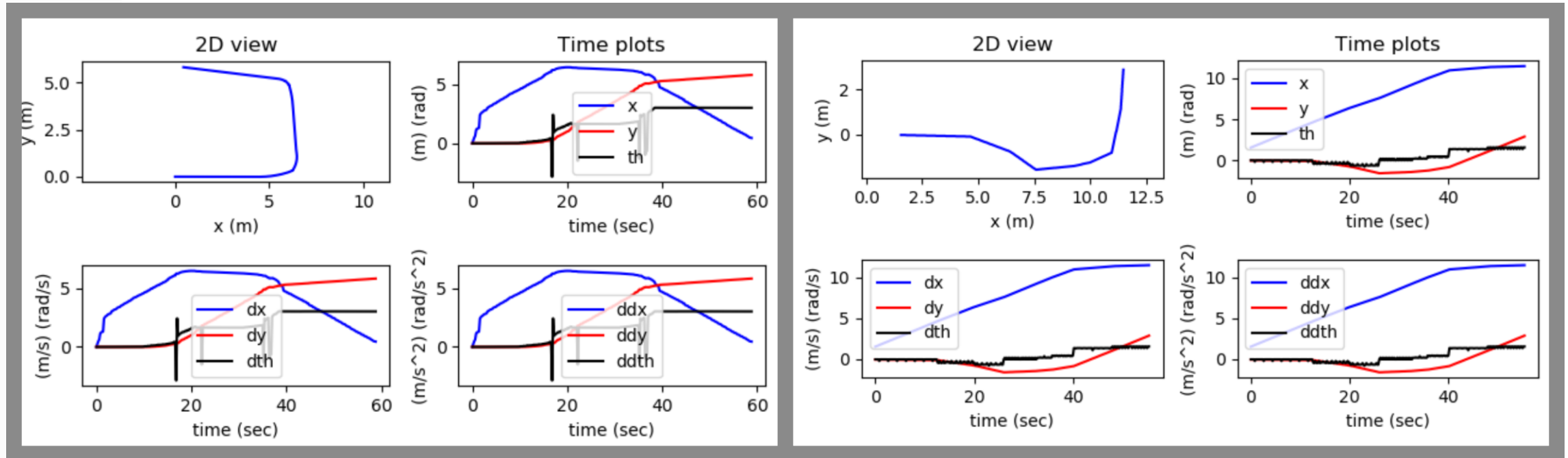
Number	Configuration
Config 1	Wheel Odometry- No EKF
Config 2	Wheel Odometry, IMU, and Magnetometer
Config 3	GPS, Wheel Odometry, IMU, and Magnetometer
Config 4	GPS and IMU
Config 5	GPS and Magnetometer

Trajectory	GPS	IMU	Magnetometer	Wheel Odometry	RMSE
Line	-	-	-	X	4,3747
	X	X	X	X	0,5341
	-	X	X	-	4,2541
	-	X	X	X	2,9828
	X	-	X	-	2,9955
L-Shaped Figure	-	-	-	X	4,7371
	X	X	X	X	0,0807
	-	X	X	-	5,9359
	-	X	X	X	1,6048
	X	-	X	-	6,9564
Ellipse	-	-	-	X	5,6842
	X	X	X	X	0,0449
	-	X	X	-	4,3991
	-	X	X	X	1,8039
	X	-	X	-	11,523
8-Figure	-	-	-	X	5,6655
	X	X	X	X	0,1559
	-	X	X	-	3,9856
	-	X	X	X	1,4785
	X	-	X	-	10,8523

Simulation Works: Cooperative Mission 1

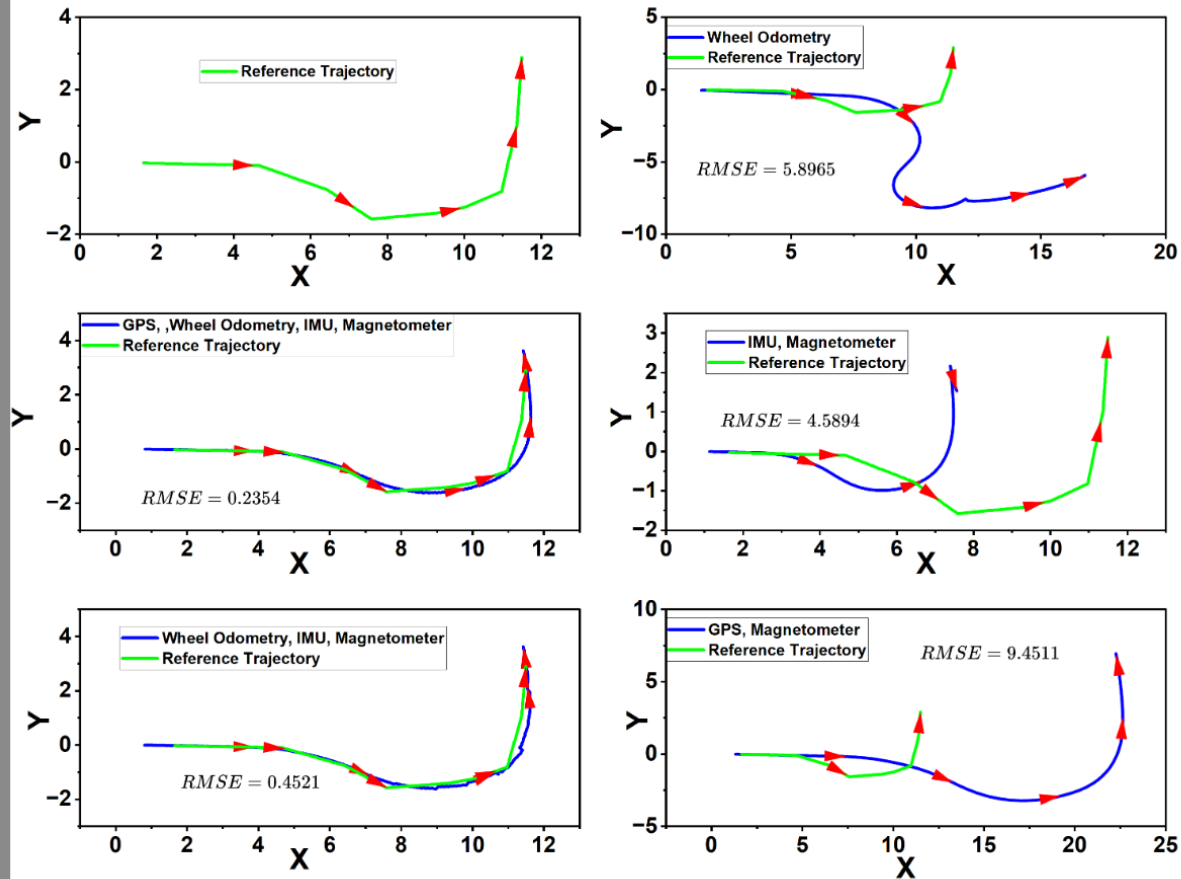


Simulation Works: Cooperative Mission 1

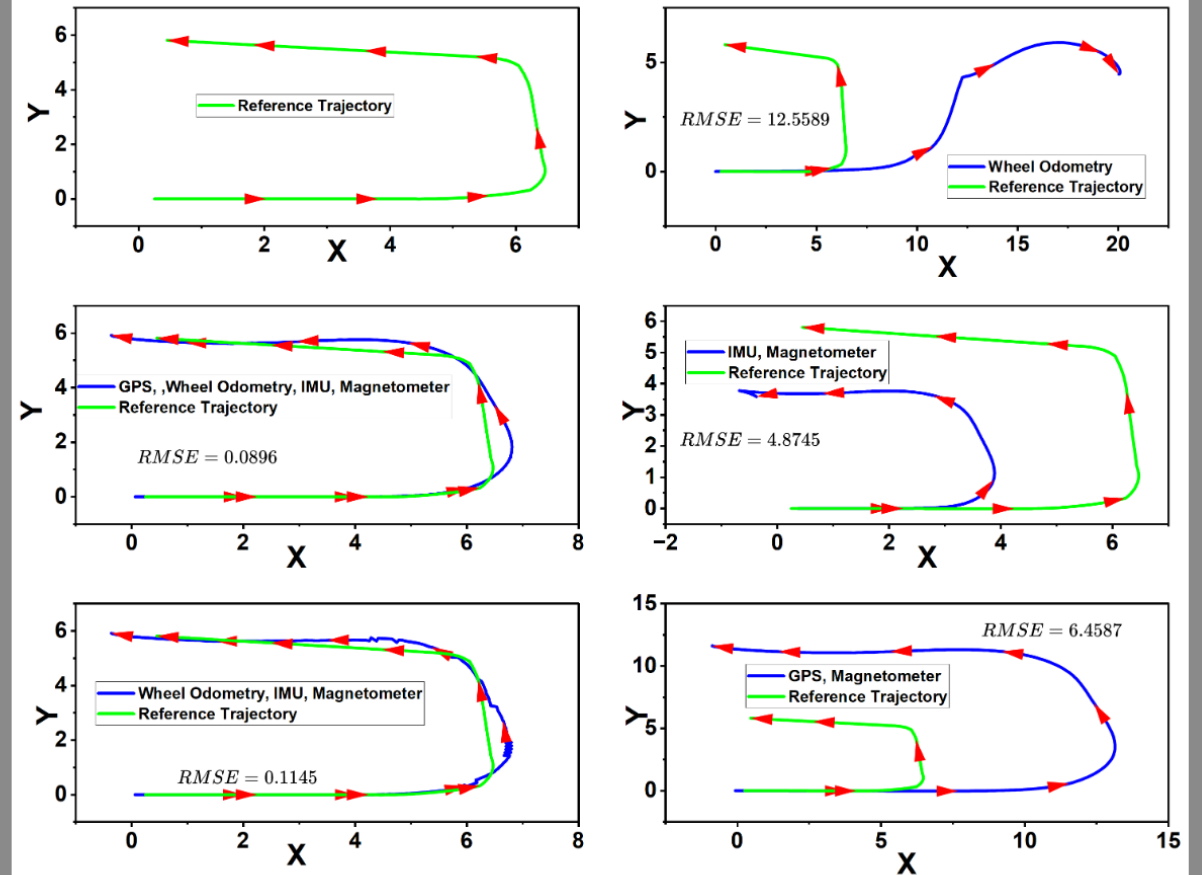


Simulation Works: Cooperative Mission 1

Localization Plots for The UGV: Mission 1-1st Trajectory



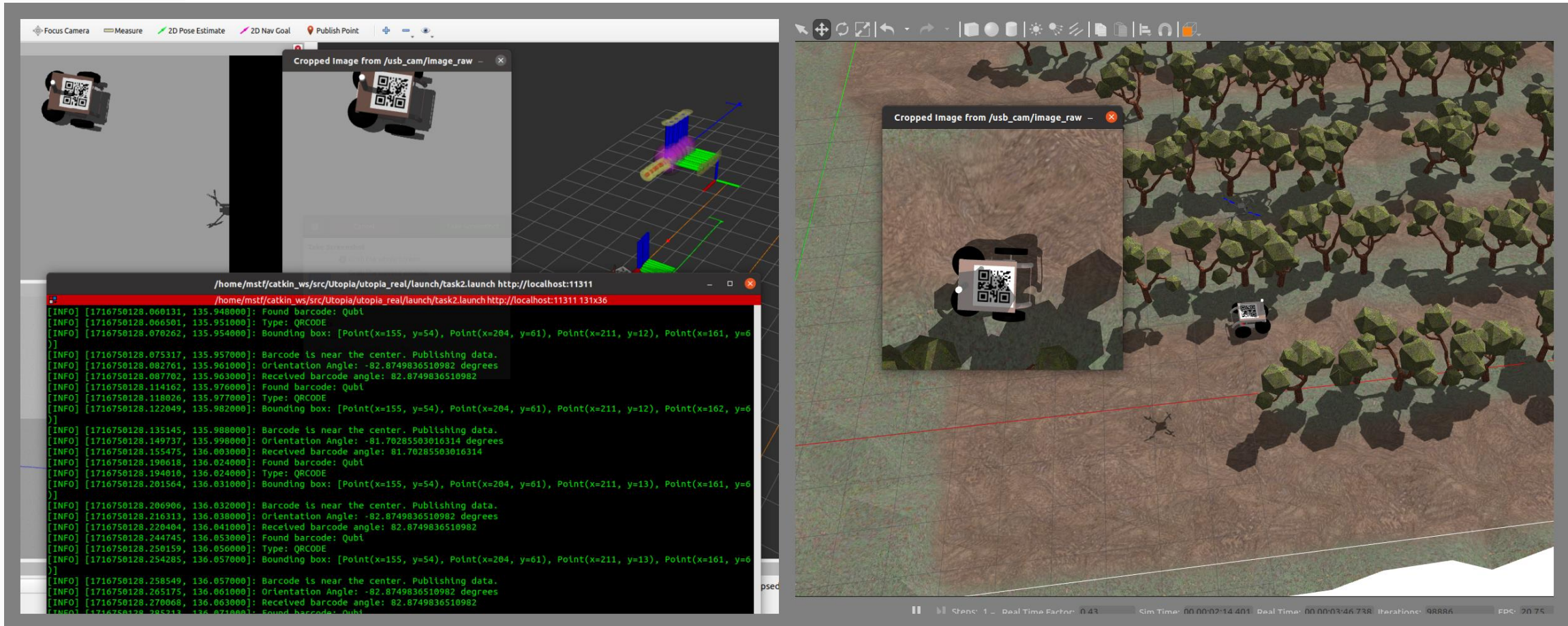
Localization Plots for The UGV: Mission 1-2nd Trajectory



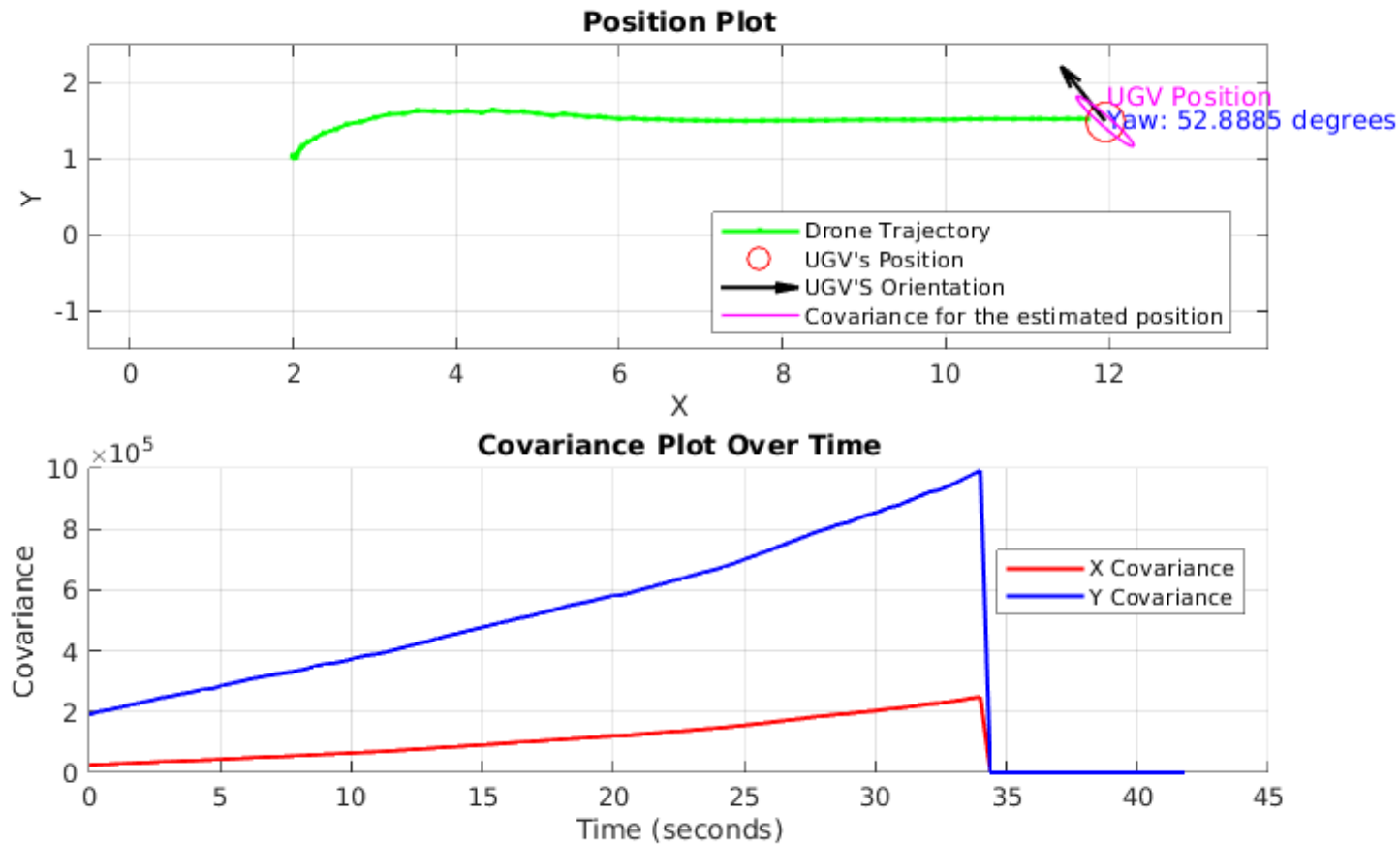
Simulation Works: Cooperative Mission 1

Trajectory	GPS	IMU	Magnetometer	Wheel Odometry	RMSE
1	-	-	-	X	5,8965
	X	X	X	X	0,2354
	-	X	X	-	4,5894
	-	X	X	X	0,4521
	X	-	X	-	9,4511
2	-	-	-	X	12,5589
	X	X	X	X	0,0896
	-	X	X	-	4,8745
	-	X	X	X	0,1145
	X	-	X	-	6,4587

Simulation Works: Cooperative Mission 2

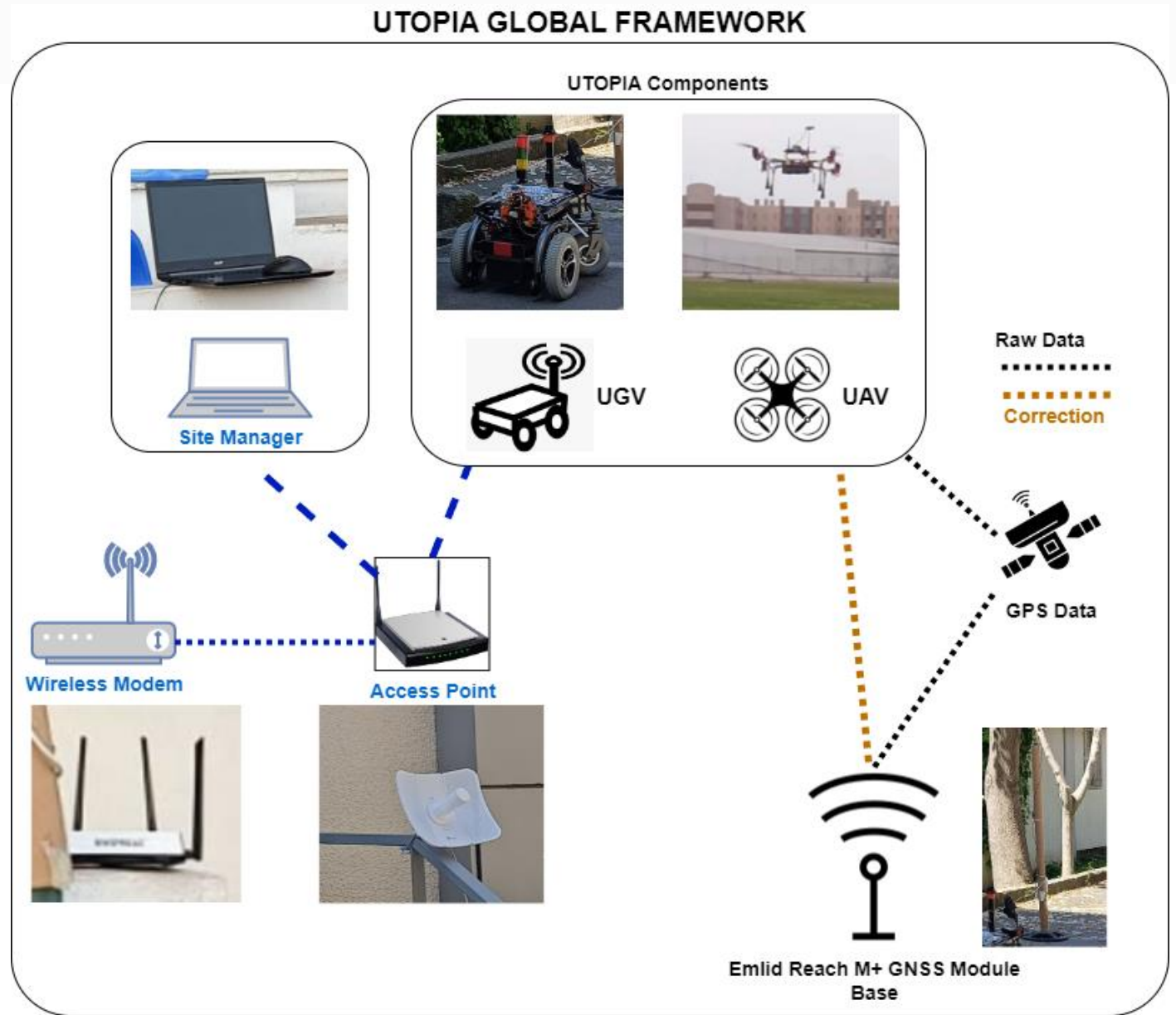


Simulation Works: Cooperative Mission 2

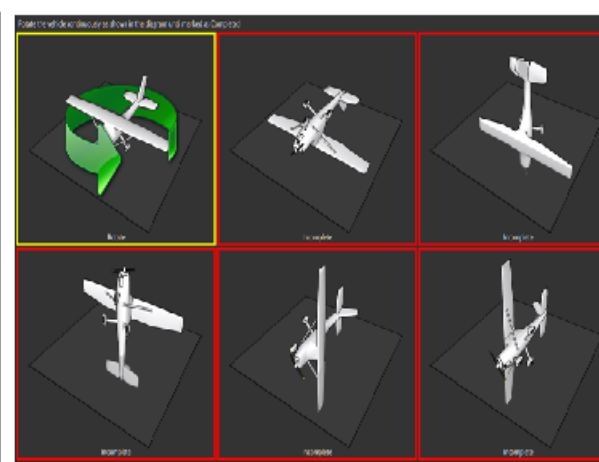
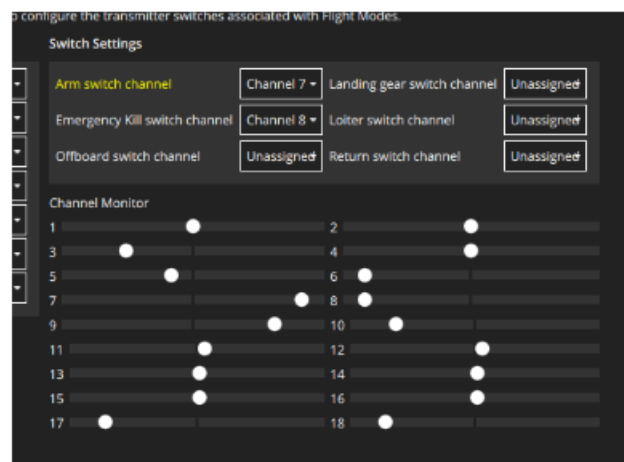


Experimental Works

Number	Configuration
Config 1	Visual Odometry- No EKF
Config 2	GPS, Visual Odometry, IMU, and Pose
Config 3	GPS, IMU, and Pose
Config 4	GPS and IMU
Config 5	Visual Odometry, IMU, and Pose

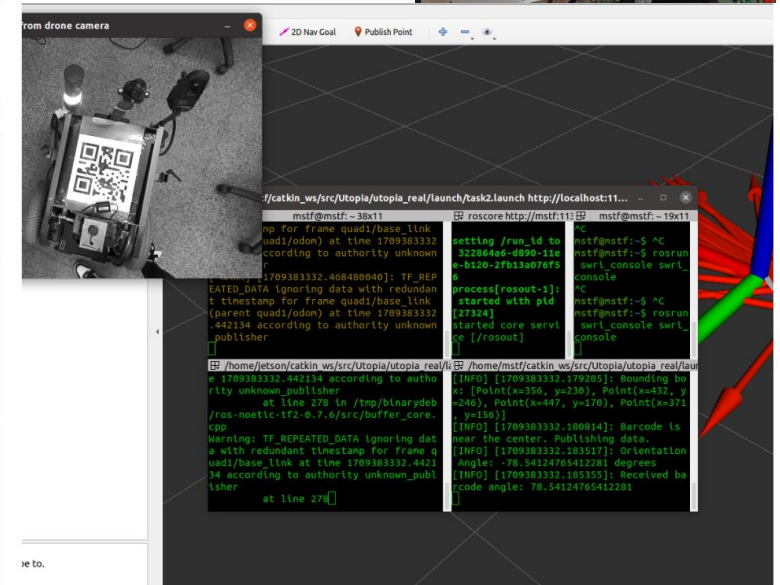
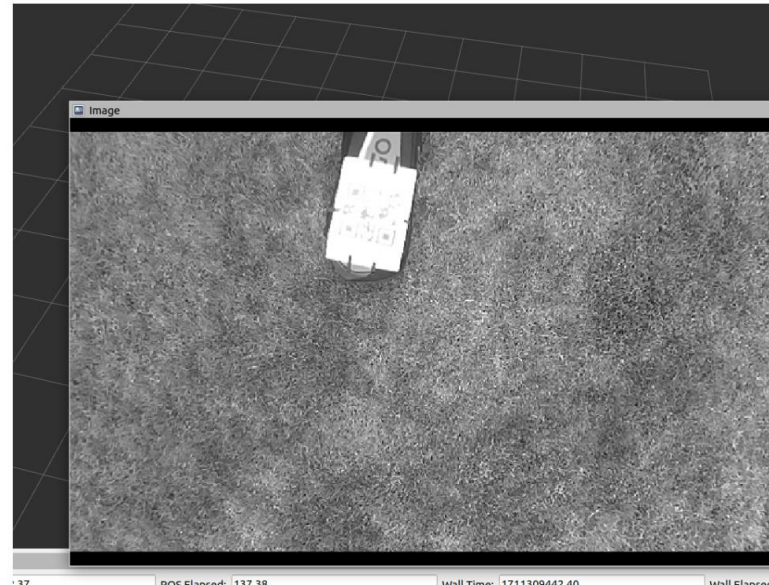


Experimental Works: UAV Calibration



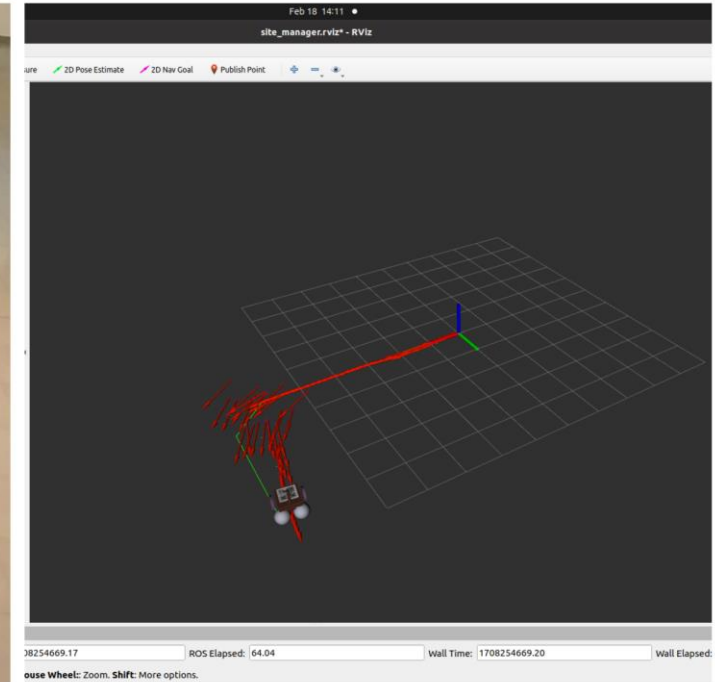
Experimental Works: UAV Calibration



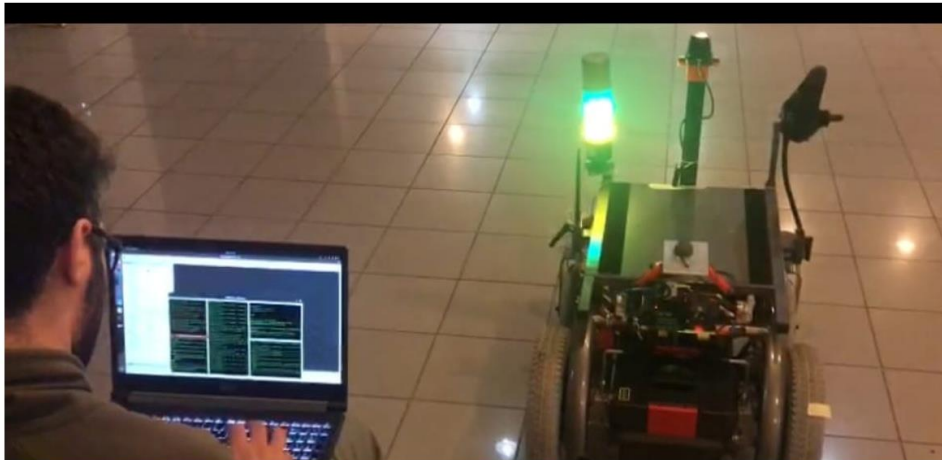
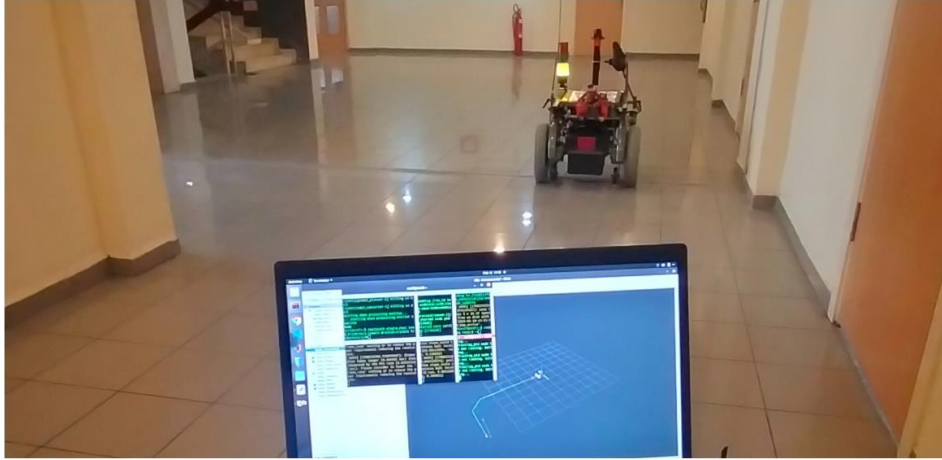


Experimental Works: UGV Calibration

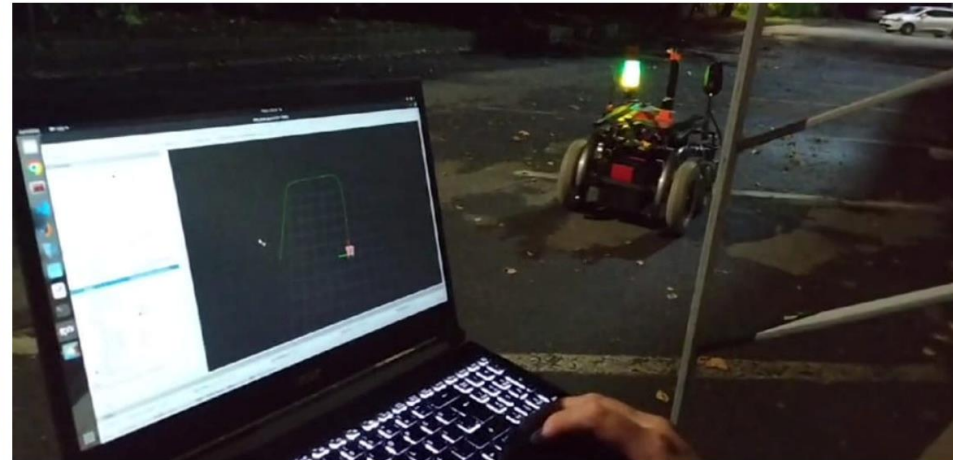
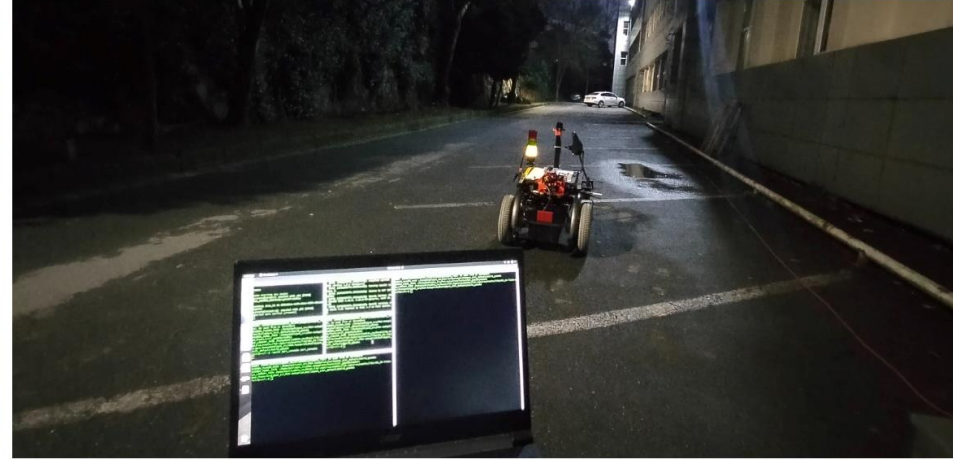
Parameter	Longitudinal	Lateral	Angular
K_p	0.4	-0.05	-0.2
K_i	0.01	-0.01	-0.01
K_d	0.05	-0.01	-0.08



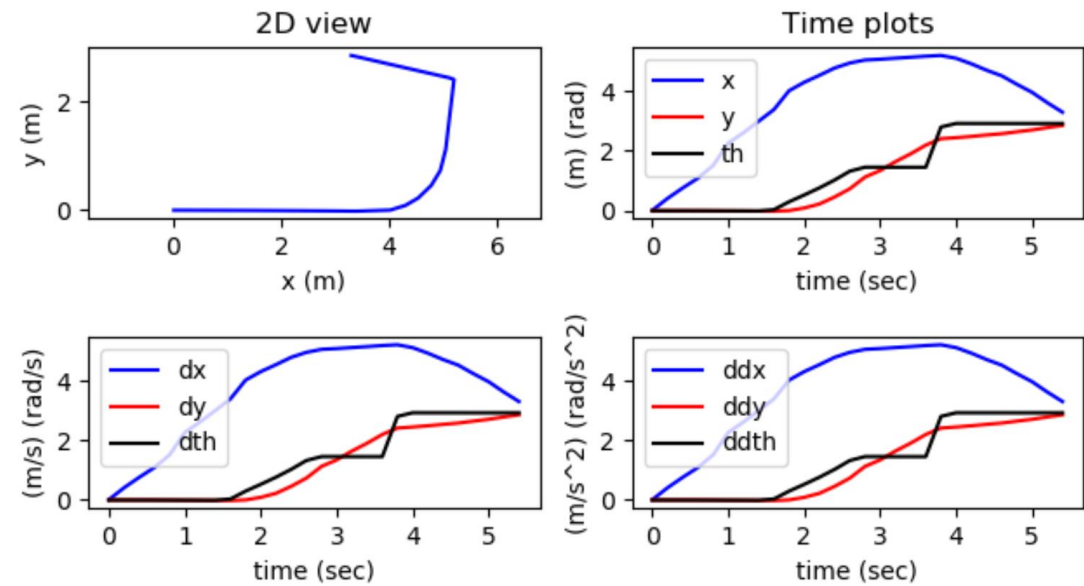
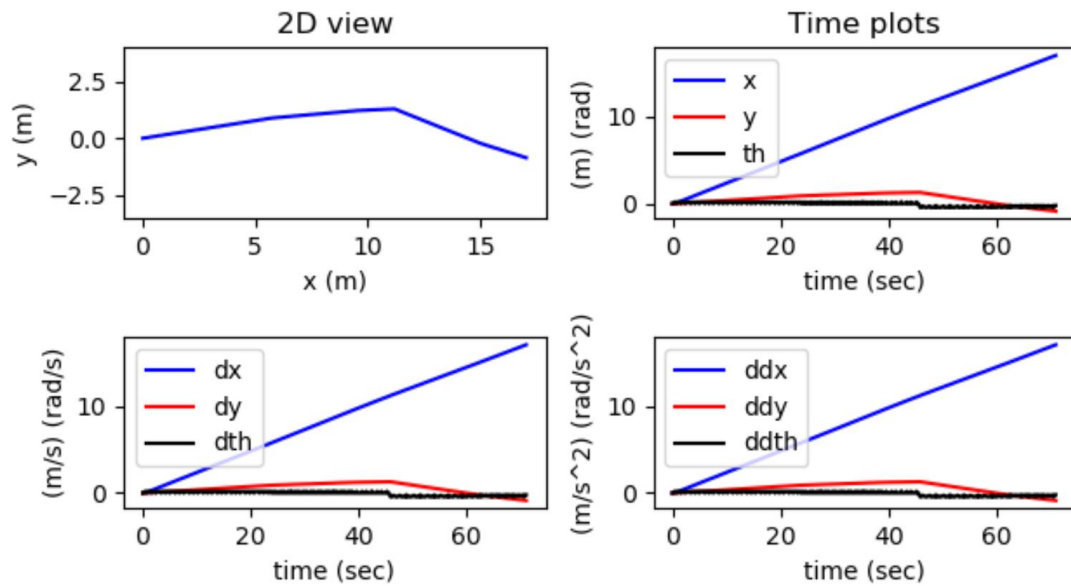
Experimental Works: UGV Calibration



Experimental Works: UGV Calibration

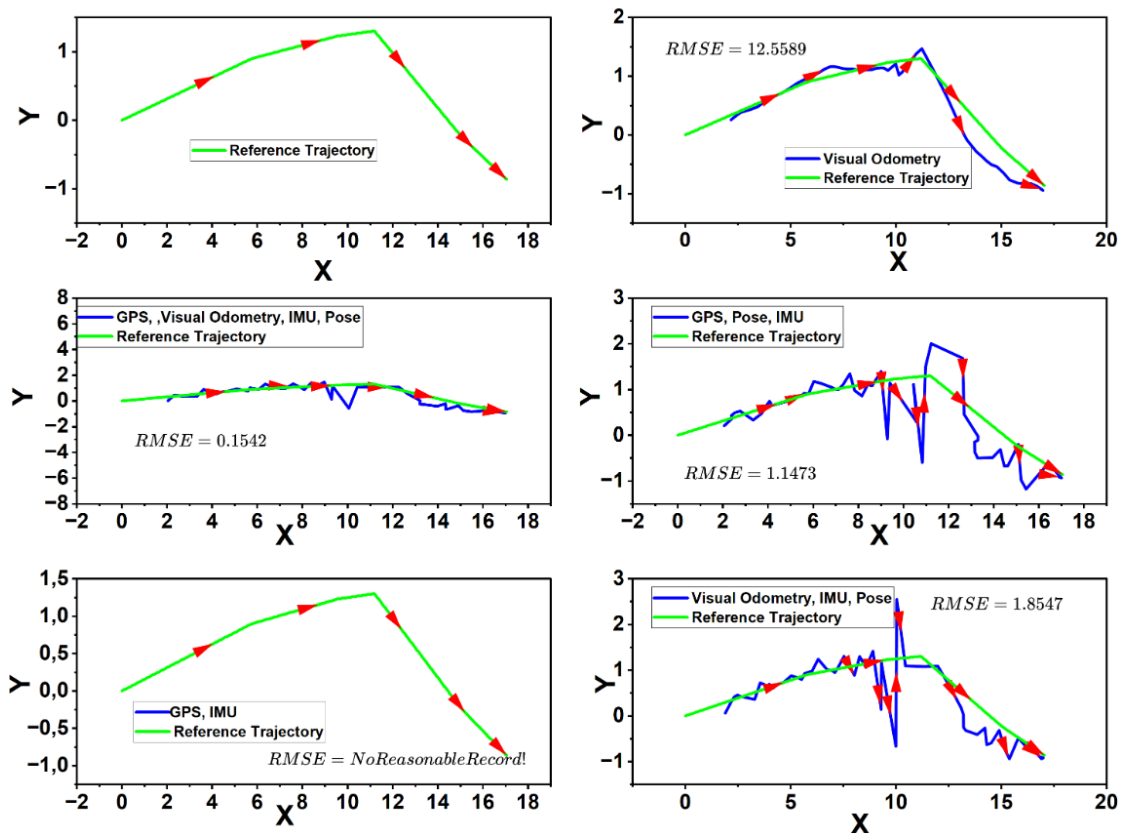


Experimental Works: Cooperative Mission 1

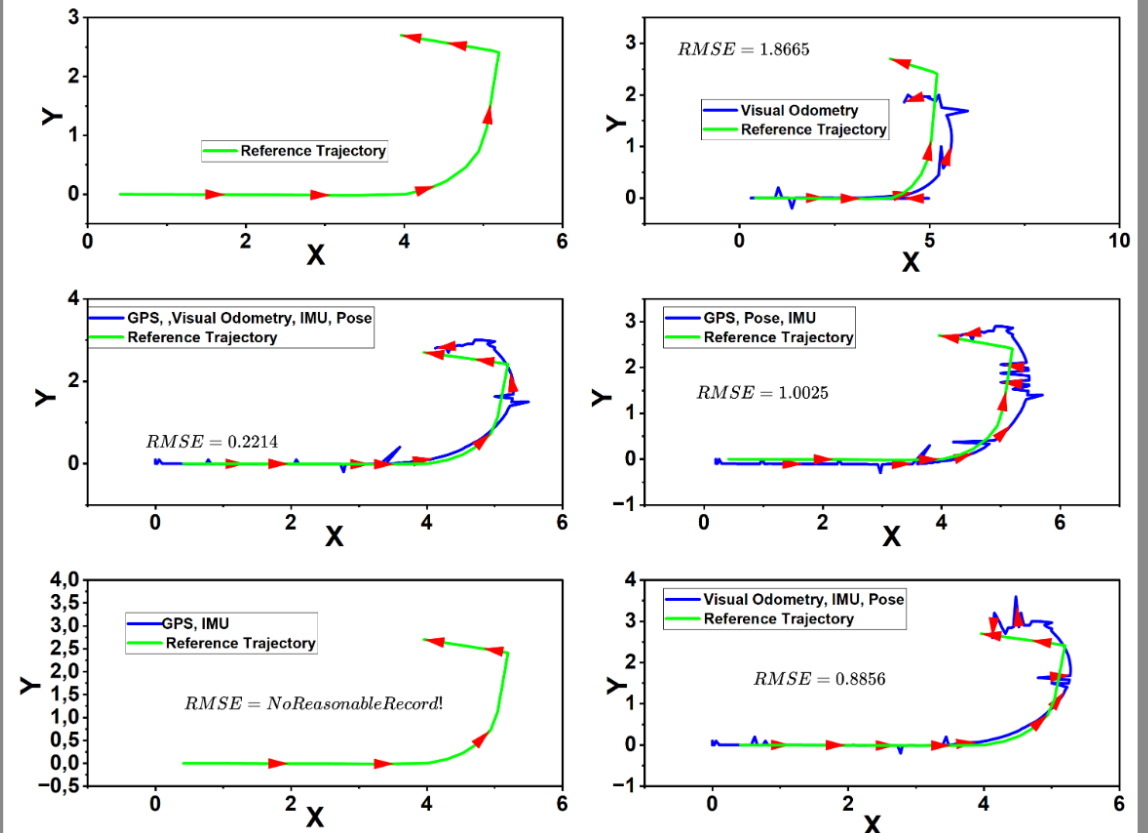


Experimental Works: Cooperative Mission 1

Localization Plots for The UGV: Mission 1-1st Trajectory



Localization Plots for The UGV: Mission 1-2nd Trajectory

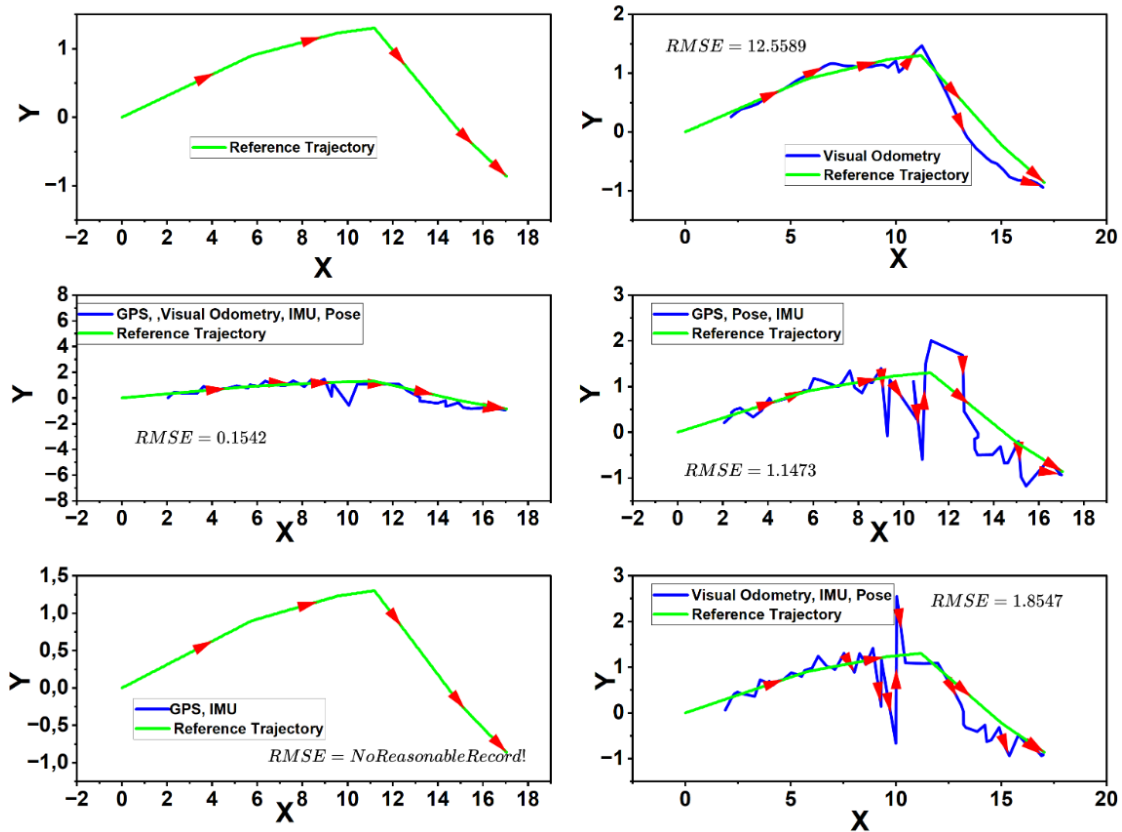


Experimental Works: Cooperative Mission 1

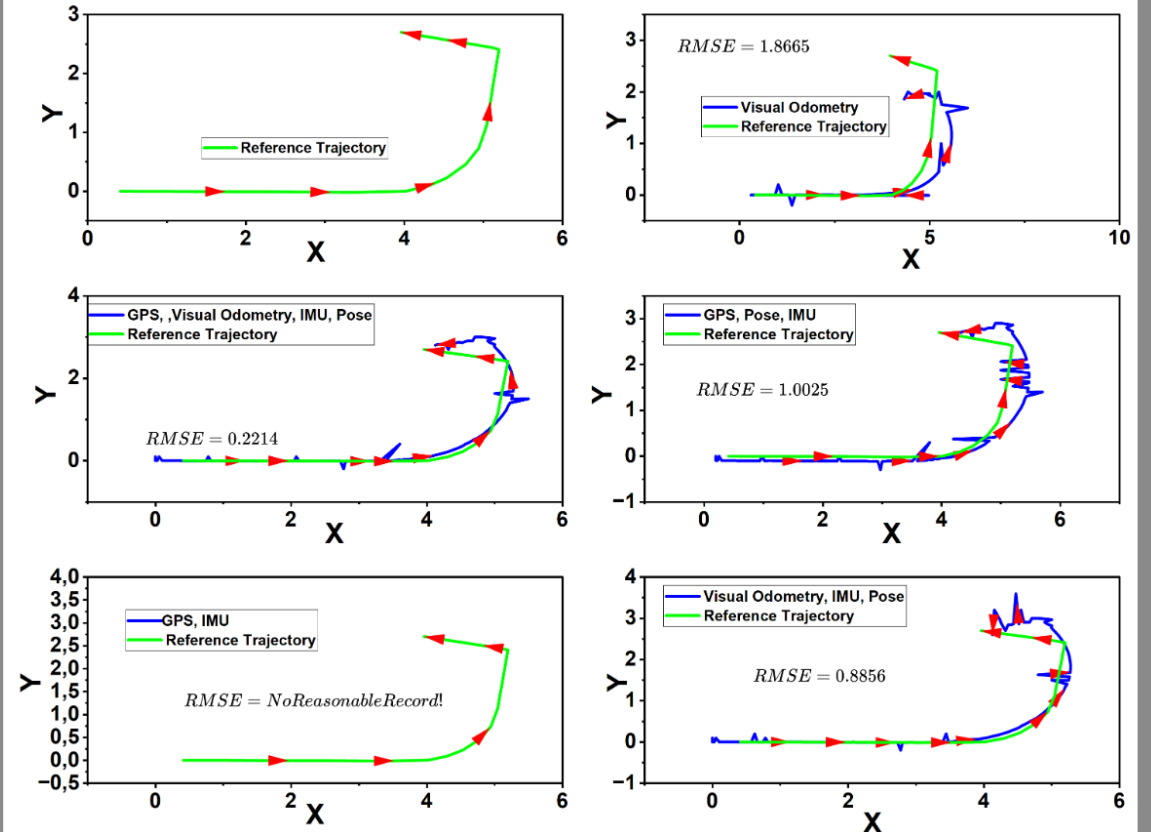
Trajectory	GPS	IMU	Pose	Visual Odometry	RMSE
1	-	-	-	X	12,5589
	X	X	X	X	0,1542
	X	X	X	-	1,1473
	X	X	-	-	NONE
	-	X	X	X	1,8547
2	-	-	-	X	1,8665
	X	X	X	X	0,2214
	X	X	X	-	1,0025
	X	X	-	-	NONE
	-	X	X	X	0,8856

Experimental Works: Cooperative Mission 2

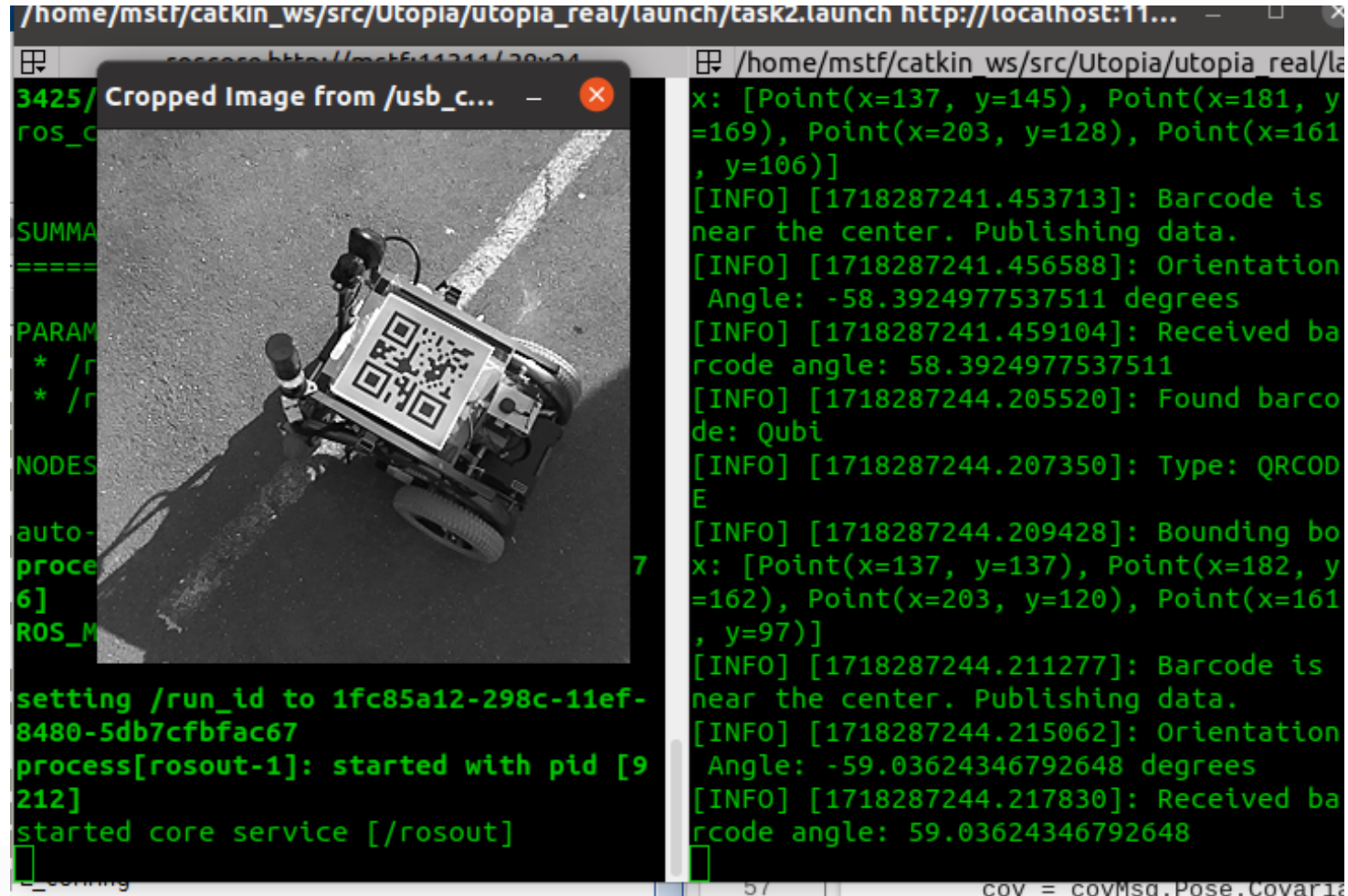
Localization Plots for The UGV: Mission 1-1st Trajectory



Localization Plots for The UGV: Mission 1-2nd Trajectory



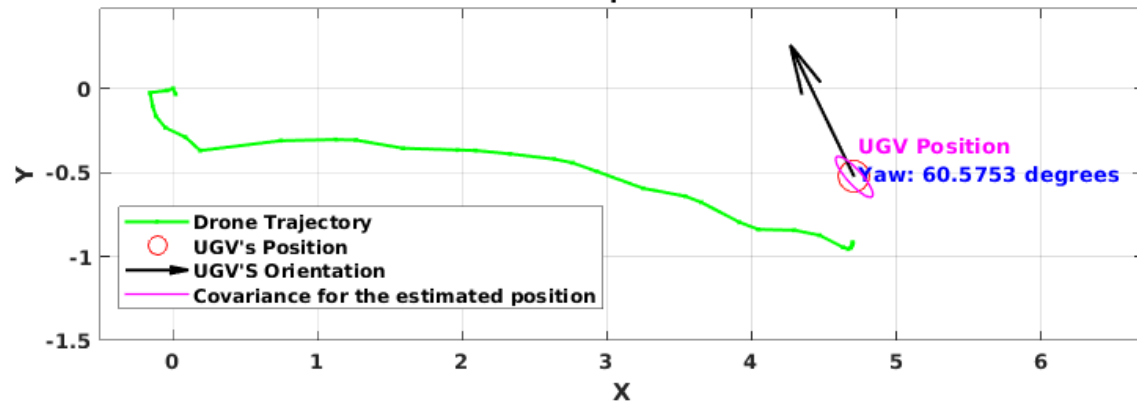
Experimental Works: Cooperative Mission 2



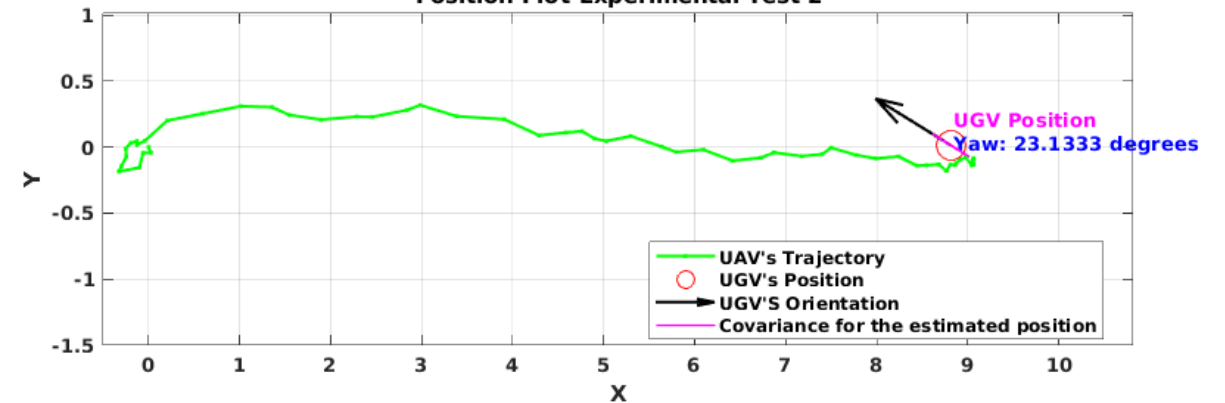
The image shows a Linux desktop environment with a terminal window, a file manager window, and a Vokoscreen 2.5.0 application window. The terminal window is the primary focus, displaying the output of a ROS2 launch command. It shows the process of starting a ROS2 master, launching a robot localization node, and then launching a quadcopter task. The logs include information about the master's startup, the localization node's parameters, and the quadcopter task's launch process. The file manager window shows the contents of the /home/jetson/catkin_ws/src/Utopia/utopia directory. The Vokoscreen 2.5.0 application window is in the foreground, displaying a recording interface with various settings and a countdown timer.

Experimental Works: Cooperative Mission 2

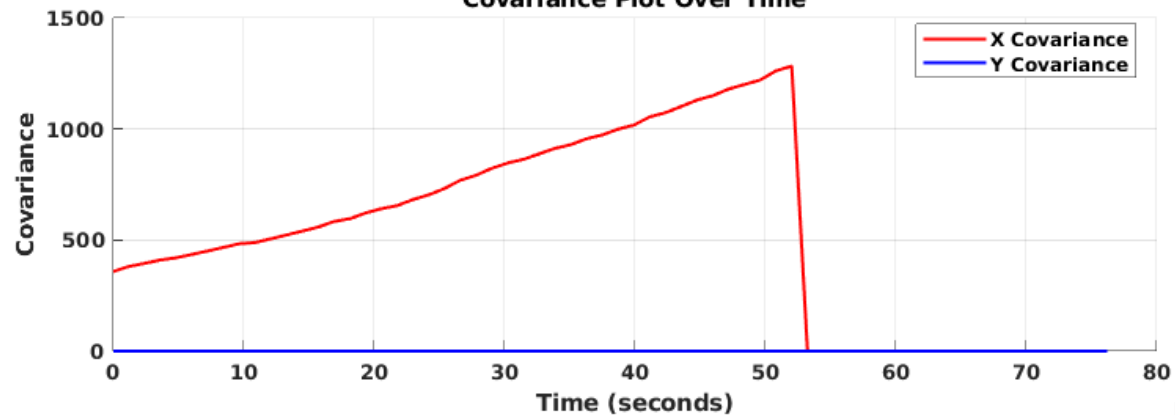
Position Plot-Experimental Test 1



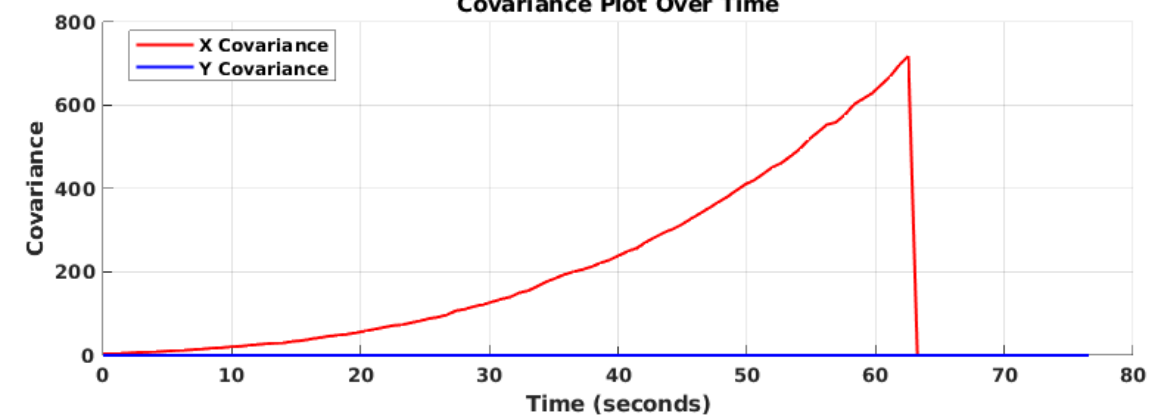
Position Plot-Experimental Test 2



Covariance Plot Over Time



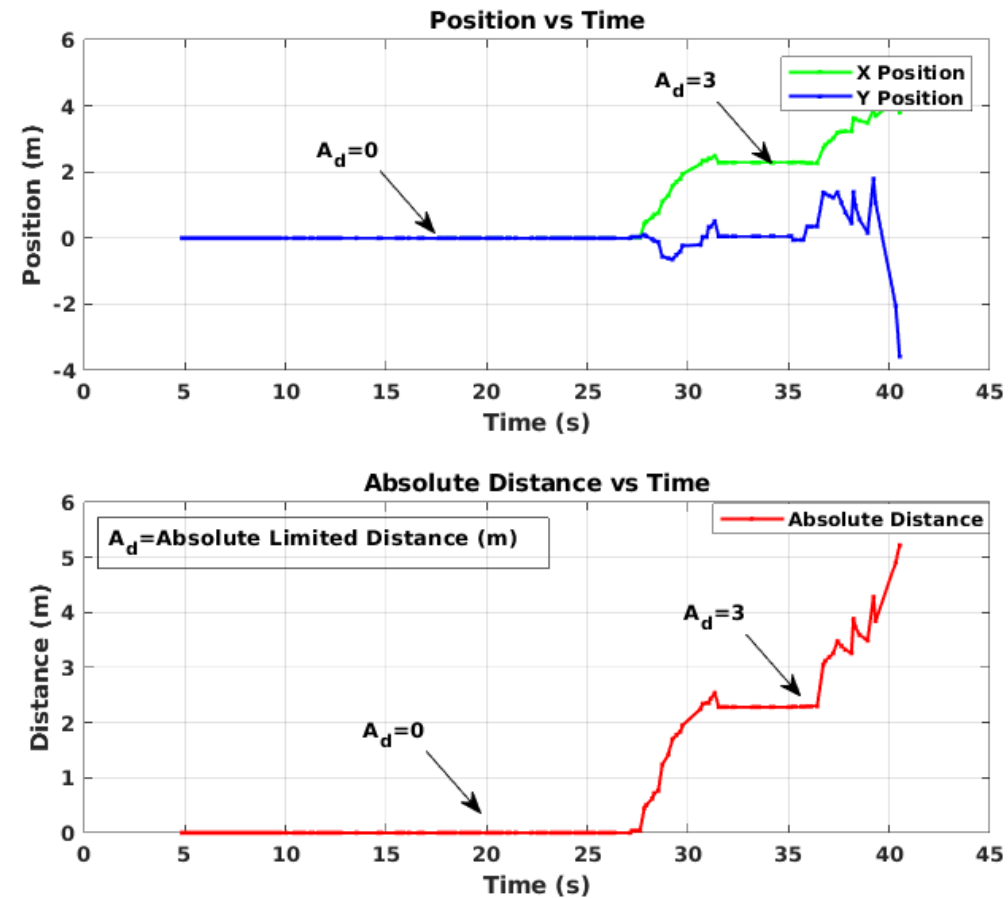
Covariance Plot Over Time



Experimental Works: Cooperative Mission 2

Test	Estimated Position		Estimated Orientation	
	X	Y	Yaw (Radians)	Angle (Degrees)
1	4,70970479	-0,5258043	1,05723771	60,5752589
2	8,81632224	0,01362048	0,40375257	23,1333182
Test	Covariance Before Correction		Covariance After Correction	
	X	Y	X	Y
1	1282,77629	3,62E-16	0,00299949	-6,45E-13
2	718,536498	1,93E-17	0,0044209	2,43E-05

Experimental Works: Cooperative Mission 3



Conclusion

Results show that adding more sensor data into the EKF lowers the RMSE, enhancing localization.

While homogeneous robot systems can achieve similar RMSE for each robot, heterogeneous systems require careful handling of sensor variances for strong localization in cooperative tasks.

Time synchronization between robots was a problem during tests. While similar robots can handle this, it affects performance in varied systems, causing issues like incorrect position estimations.

To fix this, we implemented the "chrony sync time" package for Ubuntu and adjusted the ROS environment.

Conclusion

Testing revealed issues with high computational costs and delayed messages due to outdoor wireless communication.

The variety of sensors and high-frequency data slowed the controller boards, causing localization problems.

Solutions included running the robots headless, reducing GPU use, optimizing the EKF nodes, limiting camera raw topics and image processing, and shielding sensors from magnetic effects.

Conclusion

In conclusion, this study highlights the need for diverse sensor data to enhance localization accuracy in varied robotic systems.

It aids in devising robust localization and control strategies, improving mobile robots' performance in complex settings.

Future work should focus on refining these strategies for better performance and reliability in different robotic applications.

Thank You

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